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NSP/004/041/001 Specification for the renovation of existing LV O/H Lines

1. Purpose

The purpose of this document is to provide a means of detailing the approved techniques and arrangements that should be utilised when either replacing an existing open wire network with ABC conductor or renovating the existing open wire system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/041/001	3.0 (Oct 09)	Specification for the renovation of existing LV O/H Lines
NSP/004/041/001	2.0 (Jan 07)	Specification for the renovation of existing LV O/H Lines
NSP/004/041/001	1.0 (May 06)	Specification for the renovation of existing LV O/H Lines
DSS/004/041	1.1	Specification for LV ABC Overhead Lines in Yorkshire Electricity
O.305	Feb 1998	Low Voltage Overhead Distribution Systems

2. Scope

This document details the installation requirements associated with the replacement of an existing open wire LV network with ABC conductors or retaining/adding additional conductors onto an existing open wire network.

New LV ABC Networks shall be constructed in accordance with the parent document NSP/004/041.

New overhead line services and new or replacement surface wiring shall be carried out in accordance with this document and NSP/004/043.

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3. Technical Specification

3.1 Historical Designs

Historical Designs specifications used within CE Electric UK

Design	Construction period	Types of Cond Used
BEBS L1 (9" Spacing Open Wire)	1962 - 1970	.025" HDBC / 16mm ² HDBC
ENA TS 43-30 (12" Spacing Open Wire Networks)	1970 - 1988	.05" HDBC / 32mm ² HDBC .1" HDBC / 70mm ² HDBC .15" HDBC / 100mm ² HDBC 50mm ² Plain Aluminium (Ant) 100mm ² Plain Aluminium (Wasp)
ENA TS 43-12 ABC Networks	1988 - Present	2 x 35mm ² ABC 4 x 35mm ² ABC 4 x 50mm ² ABC 4 x 70mm ² ABC 4 x 95mm ² ABC 4 x 120mm ² ABC

3.1.1 Historical Design Criteria

Historical LV networks were designed in accordance with the "Electricity (Overhead Line) Regulations 1970 which resulted in the following design criteria:

Loading Conditions:

- 380n/m² wind pressure acting on the conductors augmented with 9.5mm diametric ice.
- The minimum factor of safety on the calculated conductor breaking loads were 2.0 for open wire networks and 2.5 for ABC networks.
- MWT' Limits of 4.45kN and 6.0kN were imposed on 32mm² and 70mm² HDBC open wire lines respectively to enable extensions to be carried out to the older BEBS L1 design specification without the need for out of balance stays.
- The MWT's of the ABC lines were based on the design tables listed in ENA TS 43-12.
- All supports were designed to achieve minimum factors of safety of 2.5

3.2 Renovation of existing LV networks

The prime objective of renovating LV networks must always be to ensure that they comply with all current legislation, have all defective components removed and that any issue likely to affect the continuity of supply has been corrected or removed.

During the period 2009/10 all LV lines in CE Electric UK were subjected to a detailed condition assessment. The results from these surveys shall be used in conjunction with the following guidelines to determine how the network shall be renovated.

Appendix 14 details a table of the condition points used for the line assessment together with guidance on those "CR" ratings that must be resolved.

Appendix 15 provides a template to summarise the condition points recorded in the original line assessment so that they can be confirmed by the secondary condition survey detailed in clause 3.2.1 and used as the basis for line authorisation.

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The necessary renovation work required on each network can be categorised as follows:-

- a) Extensive Maintenance
- b) Conversion to ABC

3.2.1 Condition Assessment Survey

Despite the fact that the network has been previously inspected as part of the overall CE Electric UK asset condition survey, all networks shall be subjected to a re-inspection survey prior to actual work commencement, as the condition of the asset may have changed from its original inspection by the time work is due to commence. The inspection shall include the following:

- a) The examination of all poles
- b) The examination of all stays
- c) The examination of all insulators and binders
- d) The examination of all services and service spans for ground clearance
- e) The examination of cable terminations and auxiliary pole equipment
- f) The examination of network conductor condition
- g) The identification of any problems of proximity to buildings or trees
- h) Confirmation of adequate ground clearance

The level of detail involved in the re-inspection survey will be dependant upon the accuracy of the original survey data and the period of time between the original line survey and work starting, as in some cases this could be many years.

3.3 Extensive Maintenance

It is envisaged that this category will be applicable where less than 50% of the total number of poles on a section of network require replacement and the following criteria can be complied with:-

Note:-

To avoid confusion the phrase "section of network" shall be deemed to refer to a complete LV feeder supplied up to its normal open point.

- a) The section of network utilises copper conductors with adequate CSA for network load and voltage drop requirements and does not have any evidence of previous conductor clashing or repairs. (aluminium network conductors shall be replaced with ABC)
- b) The network Neutral conductor is the same CSA as the Phase conductors.
- c) The section of network is located a minimum of 3.0m away from the surface of any building. See clause 3.6.1 "Permanent LV Shrouding" for exceptions to this policy.
- d) The section of network is located a minimum of 3.0m away from trees capable of supporting a ladder/climber or 0.8m away from trees unable to support a ladder/climber. See clause 3.6.1 "Permanent LV Shrouding" for exceptions to this policy
- e) All other min clearances detailed in NSP/004/011 are compliant.
- f) The existing phase to phase spacing is 300mm.

The aim of carrying work out in this category is to return the line to its original design performance by means of replacement of components on a like for like basis (or their modern equivalent). It will normally involve carrying out the minimum amount of work to ensure the network components have a

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"fit for purpose condition" or health condition of 3 for a minimum life of 10-15 years as all lines will be subjected to a full re-inspection on a 10 year cycle.

Apart from the replacement of a rotten pole this work will typically include the replacement of individual line components e.g. inadequate stays, broken or obsolete insulators and ties, and replacing any mechanical connectors with compression connectors. For more details see clause 3.6

As a general rule all components classified by the inspection process as a "CR4" will be replaced. In addition any pole classified as a "CR3" including steel poles will also be replaced.

3.4 Conversion to ABC

Where >50% of the total number of poles on a section of network require replacement or any of the above conditions cannot be complied with, then the network shall be converted to ABC.

Additionally ABC shall be used for the following:

- Construction or extension of all new LV overhead lines
- As a means of providing supplies formally achieved via surface wiring or eaves mains see NSP/004/043.
- In areas identified as high risk by DSS/023 "Risk Assessment of Overhead Lines"
- Area's identified in NSP/004/011 "Guidance on Overhead Line Clearances" as being not suitable for "non effectively insulated conductors" or "Permanent LV Shrouding"

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3.5 Refurbishing LV Networks with ABC conductor

3.5.1 Conductors, clearances & conductor design ratings

Replacement conductors shall be as detailed for new construction in NSP/001/041

Main Lines : 4 x 120mm² or 4 x 70mm² (with and without protective conductors on 4 x 120mm²)
Service Lines: 4 x 35mm² or 2 x 35mm²

Additional information has been included in NSP/004/041 to cover for work or extensions to existing 4 x 50 & 4 x 95mm² sizes.

The minimum height of ABC networks above ground level and it's clearances to other objects shall be compliant with NSP/004/011.

Historically lines were designed to operate at 50°C. Where possible renovated lines shall provide max operating temperatures and clearances to allow operation at 75°C in accordance with new construction. However poles shall not be replaced solely to provide an increase in existing operating temperature providing the existing minimum ground clearances at the original max design temperatures still comply with NSP/004/011.

3.5.2 Design basis of refurbished networks

When a line has been renovated to ABC it shall be deemed to have performance levels equivalent to newly constructed lines. See NSP/004/041 for design details. (Where a line undergoes extensive maintenance it shall be deemed to have performance levels as detailed in clause 3.1.1.)

3.5.3 Wood poles - General

Existing wood poles may be retained provided they are of minimum classification Medium to BS1990 and that they have been examined and tested in accordance with NSP/004/112. (Appendix 12 has been included to assist in the identification of existing pole grades).

Retained poles shall be capable of providing continued service for a minimum of 10 years. Supports that have been identified with minor deterioration but have been confirmed as suitable for continued use shall only be retained when they do not carry additional pole mounted plant. Eg cable terminations.

Where poles have been inspected and confirmed to be suitable for continued use but do not have a 3m gouge mark to indicate the correct installed sinking depth, they may be retained providing they are situated in locations that can be safely accessed using mobile access platforms or where the pole can be temporarily stayed in four directions. Where this is not the case the pole must be replaced.

All supports that are retained and are in excess of 30 years old shall be treated with BORON ROD preservative.

Existing poles which are retained shall be fabricated with a new 22 mm diameter hole, 300 mm from the pole top; unused existing holes must be plugged using appropriate size wood plugs as indicated Drawing No. 1091010247. Drawing 1091193302 provides details on converting an existing openwire pole to carry ABC.

Replacement poles shall be of a minimum medium grade and treated with water soluble preservatives. New creosoted poles are not allowed on LV Networks. The poles shall be fabricated to Drawing Nos. 1091193301 sht4.. Care shall be taken when selecting replacement poles to take account of the increased pole sinking depths and the potential reduction in ground clearances.

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Attention shall also be paid to ensure that the optimum position is selected for the new pole and that it is not simply located adjoining the existing pole or in such a position that it results in the new pole being placed out of alignment.

Appendix 1 provides details of arrangement drawings to allow conversion of open wire networks to ABC.

3.5.3.1 Pole Foundations

All foundations and sinking depths shall be as detailed for new construction. See NSP/004/041 clause 3.4.1 for pole sinking depths.

3.5.3.2 Pole Caps

All wood supports shall be either fabricated with a 30° four quartered apex or be fitted with a suitable form of pole cap.

3.5.3.3 Un-stayed angles on intermediate supports

Un-stayed angle supports shall be avoided wherever possible. However they are permitted providing they comply with the following criteria:

ABC Main – On Medium grade poles the max angle of deviation does not exceed 7°

ABC Main - On Stout grade poles the max angle of deviation does not exceed 19°

Extended Service Span - Medium grade pole with max angle of deviation 30°

In all cases the max pole height is limited to 12m and the poles must be sunk to a depth of 1.8m and fitted with 2 x ENA TS 43-91 Type 2 (1300 x 250 x 125)mm blocks located 500mm below ground level and attached with a M20 bolts. Where access and ground types are suitable, augured foundations may be used as an alternative to blocked foundations. See drawing 1090431216 for foundation details.

Tubular steel poles of minimum class 'K' may be utilised to provide un-stayed angles in excess of the above limits. Their use should be limited to a last resort due to the onerous cost and protracted installation requirements.

Where existing un-stayed supports use offset brackets to minimise the angle of deviation on the pole in accordance with Drawing No. 1091193309 they may be retained, providing they are in sound condition.

The brackets shall have the top insulator fixing hole redrilled to allow the attachment of an M20 hook nut (Drawing No. 1000431401, sheet 2).

3.5.3.4 Stayed Supports

Stayed supports shall typically be of medium grade and be designed to be in accordance with the minimum stay spread tables detailed in NSP/004/041 Appendix 9.

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3.5.3.5 Section Supports

Section pole arrangements shall supported on minimum class medium grade poles for all ABC conductor sizes provided the pole is a stayed support. Details of minimum stay angles can be found in appendix 9.

Arrangement	Drawing Number
Renovation of Open Wire Angle Pole (30° dev) to ABC	1091193303 Sht. 1
Renovation of Open Wire Angle Pole (60° dev) to ABC	1091193303 Sht. 2
Renovation of Open Wire Section Angle (90° dev) to ABC	1091193304
Renovation of Intermediate Angle Support to Typical ABC Tee off	1091193306

Transition supports between existing open wire networks and ABC networks shall be arranged as shown on drawing number 1000431208. Details on the requirements for out of balance stays between the two design systems can be found in appendix 10.

3.5.3.6 Terminal Supports

Terminal arrangements shall be supported on minimum medium grade poles for all ABC conductor sizes. Details of minimum stay angles can be found in appendix 9. Terminal supports shall be arranged in accordance with the following drawing numbers.

Arrangement	Drawing Number
Renovation of Open Wire Terminal Pole to ABC	1091193307
Renovation of Open Wire Terminal Pole to ABC (C/w Cable Box)	1091193308
Renovation of Intermediate Angle Pole with Tee off to ABC	1091193305
Tee off from section support	1000431212

Tee-off poles shall be a minimum grade stout.

3.5.4 Tubular Steel Poles

Steel supports shall be both visually and ultrasonically inspected in accordance with NSP/004/112 before they are considered for continued use. The ultrasonic inspection process shall be applied to a depth of at least 80mm below ground level (deeper if it is suspected that ground levels may have changed during the life of the support).

All supports with visible holes or thickness measurements of less than 80% of the original support thickness shall be replaced. The original support thickness values are dependent upon the original steel pole class and age.

Where a new steel support is required it shall be class K in accordance with drawing no. 1091193501 sheet 2. All new K class supports are supplied with a 30 year protective galvanised coating (year of manufacture stamped on the pole).

Appendix 1 provides details of arrangement drawings to allow conversion of open wire networks to ABC.

Irrespective of the condition of a steel support, its continued existence on the LV network shall be re-assessed. In many cases the network may have evolved in such a way that the steel support could have its duty replaced by a standard wood support. Due to the higher maintenance costs associated with steel supports, where appropriate they shall be replaced with wood supports.

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In order to secure ABC to sound steel supports, the following methods may be adopted:

- Use of pre-drilled 22mm diameter holes located 300mm from the support top
- Collars, of appropriate size, to drawing 1091193510 sht2

Note

See clause 3.5.7 for special earthing and bonding requirements for steel supports.

3.5.4.1 Tubular Steel Pole - Numbering

Steel poles shall be numbered by means of stencilled and painted numbers in accordance with NSP/004/109.

3.5.4.2 Tubular Steel Pole - Allowable Angles of deviation

Allowable angles of deviation on historical un-stayed steel supports carrying ABC conductor

Location	4 x 35mm² / 50mm² ABC	4 x 70 /95/120mm² ABC
Terminal Positions		
	Type C	Type D
	Type D	Type G
	Type F	Type J
	Type G	Type K
	Type J	
	Type K	
Angle Positions		
Type B	Not recommended	Not recommended
Type C	50°	30°
Type D	80°	50°
Type E	25°	15°
Type F	65°	40°
Type G	90°	70°
Type J	90°	70°
Type K	90°	70°

See drawing no 1091193501 sht. 2 for details on the identification of historical steel pole types

3.5.4.3 Tubular Steel Pole Foundations

New tubular steel supports shall be erected on a concrete paving stone located 1675mm below ground level then surrounded with between 100-300mm of concrete depending upon the pole type. See drawing 1091193502 Sht. 1 for details.

Existing steel supports shall be protected from corrosion, at ground level, through the application of a special paint system and K10 protective coating as shown on 1091193502 Sht 2. (K steel poles are supplied complete with a protective coating)

Where existing protective casings are of the Bitulac weatherseal type, they shall be removed and cleaned prior to the application of the protective coatings. See NSP/004/103 for details.

3.5.4.4 Pole Caps

Steel pole caps, as shown on drawing no 1091193505 sht 2 must be secured to all steel supports. Except for J & K class supports which have a factory welded 6mm thick plate.

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3.5.5 Stays

Existing and replacement stays shall be compliant with the requirements of NSP/004/041 clause 3.5. Further guidance on stays can be found in NSP/004/104 and Drawing no 1091193326.

Stays shall be inspected as part of a condition survey prior to a network being renovated. Where, as a result of this survey, the existing stays are found to be in sound condition and stay spreads are within the requirements of NSP/004/041 Appendix 9, then they may be retained. Where existing multiple stays require replacing, care should be taken over the replacement requirements as ABC stay loadings are reduced in comparison with open wire configurations.

Where stays are applied to steel poles, this shall be achieved as shown on drawing no. 1091193608 and in general accordance with the provisions detailed above.

Stay insulators shall always be inserted in all stays assemblies (except for steel poles).

3.5.6 Safety Signs, labels and notices

All safety signs, labels and notices shall be complaint with NSP/004/041 and guidance document NSP/004/109.

3.5.7 Earthing & Bonding

See NSP/004/041 clause 3.11 for general details on earthing requirements for renovated ABC networks and clause 3.11.2 for specific details relating to steel poles.

3.5.8 Erection of Conductors

ABC shall be erected in accordance with the new construction design/erection charts detailed in NSP/004/041 Appendix 3 - 7.

Note. ABC erected on steel poles does not require any reduction in conductor tensions

See NSP/004/041 clause 3.6 and NSP/004/105 for more details on the approved techniques for erection of ABC conductors.

3.5.9 Services Connections

See NSP/004/041 clause 3.9.2 and NSP/004/043 "Specification for Overhead Services, Surface Wiring and Eves mains" for more details on service connections.

3.5.10 Services –Hybrid or copper concentric

Insulator attachments for existing hybrid services shall be repositioned higher up the pole to allow the conductor tails to be directly terminated into the IPC's connectors on the ABC. This will maximise ground clearances to the service span and reduce the number of overall connections in the service.

All service connections shall be as detailed in NSP/004/043 clause 3.2.1 ensuring that where possible a phase balance is maintained on the ABC main.

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3.5.11 Services - Open Wire Construction

Where PVC covered open wire services aerials are encountered they shall have their hand binds removed, the insulator repositioned higher up the pole and the service re-terminated using limited tension service preformed deadends.

The existing 16 mm² PVC insulated service conductor shall be utilised to form the connection between the ABC and the open wire service aerial. The tails shall be connected to the ABC main using single bolt IPC's drawing no. 1000431414 sht1 ensuring that the tapping conductors are secured to the bundle using nylon cable ties and dressed/fixed to the pole using cable cleats.

Where the existing 16 mm² open wire tail is of insufficient length to allow direct connection to the IPC, a short length of 16 mm² PVC insulated conductor shall be used.

For more detail on the service connections or attachment method at the building side of the service aerial see NSP/004/043.

Where LV networks are converted to ABC construction for ESQCR proximity issues, any bare neutral and Hessian covered aerials shall be replaced as they are not effectively insulated and cannot be terminated onto the ABC main without some additional forms of shrouding or insulation.

Bare aluminium service aerials shall be replaced but PVC covered aerials may be retained.

Where LV networks are converted to ABC due to a general network condition or load related issue, then it is permissible to retain bare neutral open wire services provided they comply with all other aspects detailed in NSP/004/043 and NSP/004/011. Hessian covered services shall always be replaced.

3.5.12 Services - Public Lighting

Where, prior to renovation, the public lighting was supplied using switch wires, the local lighting authority must be informed in order that each light on the network can be separately controlled by photo electric cells if they decide to retain lamps on network poles. See NSP/004/043 for details on the connections to the public lighting service cables.

3.5.13 Mains Cable Terminations

3.5.13.1 Compound Filled

Wherever practical compound filled PILC cable terminations shall be replaced by modern heat shrink terminations. This will normally be accommodated by installing a new length of waveform LV cable up the pole and installing a straight joint at the base of the pole.

However, where the termination is found to be in sound condition and there is no evidence of compound leaks or damaged conductor terminals, the cable box may be retained. In making the decision to retain an existing termination, an assessment shall be carried out on the ease of maintaining supplies to customers associated with the termination in the event of a possible future failure.

Connections to the cable box shall be made by securing each ABC 'tail' to the appropriate cable box terminals and using bimetal compression lugs in accordance with drawing no 1091193307. Where a box is mounted on an intermediate pole, a short length of ABC conductor of the same CSA as the main line shall be used to form the jumper and be connected to the ABC main using two bolt IPC's to

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drawing no. 1000431414, sht 2. Where the overhead network requires separation from the underground network for operational purposes it shall be achieved by disconnecting the lugs from the cable box.

Alternatively consideration should be given to the modification of the existing jumpers to the ABC network. In many cases it will be prudent to install a set of pole mounted LV fuses in the jumpers to create a permanent section point.

3.5.13.2 Heat Shrink Cable Termination

Heat shrink cable terminations for both PILC or Waveform cable shall be connected to the renovated ABC network generally in accordance with drawing no.1091193308. The jumper connection between the cable termination and the ABC network shall be created by continuing the ABC network directly into the new heat shrink cable terminations. Alternatively the existing copper PVC Insulated and Sheathed tails from the heat shrink termination shall be jointed onto the ABC tails using Bi-metal connectors to drawing 1091010649. Once completed all joints shall be covered with insulated sleeves.

When the cable termination is on an intermediate pole, a short length of ABC of similar CSA to the main line shall be utilised to form the jumper. This shall then be connected to the underground cable as above and to the ABC main using double bolt IPC's.

If disconnections are required from heat shrink terminations for operational purposes, this shall be achieved by cutting the ABC tails. Reconnection shall be made with non-tension connectors that must be covered with an insulating sleeve.

In locations where it is foreseen that equipment disconnections and reconnections will be required, section fuses should be installed.

3.5.14 Section Fuses

The existing section fuses shall be replaced with new fuse units with lugged connections to drawing no. 1091471530. See drawing no 1000431207 for details on the typical pole assembly drawing.

3.5.15 Connections to Static Balancers & Regulators

See NSP/004/041 clause 3.9.5 for details on connection to static balancers or regulators.

The metalwork of the of the regulator or static balancer shall be bonded to the neutral by means of a short length of 32 mm² annealed copper Black PVC (6) covered conductor.

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3.6 Refurbishment of Open Wire Networks and the addition of conductors

Renovation of existing open wire distribution systems and the erection of additional conductors shall be generally in accordance with ENA TS 43-30. This part of the specification is included to specify the design and constructional requirements where an existing section of network is not converted to ABC or where additional phase conductors are installed.

An existing open wire network will only be retained where all the provisions of clause 3.3 of this specification can be satisfied.

Note:-

Where a network has been refurbished and the existing conductors have been retained the refurbished network shall be deemed to be constructed to the original design criteria detailed in ENA TS 43-30 and Clause 3.1.1 of this specification.

3.6.1 ESQCR Proximity Issues - Use of Permanent conductor shrouding

Where a condition assessment carried out on an existing open wire LV network prior to refurbishment indicates that the existing conductors are suitable for continued use with the exception of a small number of proximity clearances to buildings or climbable trees, then consideration shall be given to the use of short lengths of permanent conductor shrouding.

Proximity clearances issues occur when non-effectively insulated conductors are located within 3.0m of a building surface, any permanent object that a ladder may be rested against or any climbable tree.

Permanent conductor shrouding consists of a 0.7mm layer of black polypropylene split tubing that is supplied in 3m lengths and applied over existing conductors to provide insulation properties to the bare conductor. The 21mm O/D size version will be standardised for use on all network conductor sizes.

Conductor shrouding shall only be considered for use on an occasional span basis and for use where the extent of the proximity infringement is less than a span length or extends to a point into the span that allows the shrouding to provide insulation without the need to apply additional insulation onto pole fittings and connectors. Whilst this system may be suitable for protecting the gable end of a house or a large climbable tree from an adjacent LV network, it is not practicable for use in protecting a complete street running parallel to a network.

Where the design of the network would result in the need for multiple spans to be protected the network shall be converted to ABC.

Note

Effectively insulated conductors must maintain a min of 1.0m clearance from a building surface unless otherwise varied by clause 3.6.3.2 of NSP/004/011.

3.6.2 Conductors

The condition of the existing conductors shall be inspected for signs of damage or previous conductor clashing. Where conductors are required to be replaced, the network must be converted to ABC

Where the conductors are in good condition and compliant with all conditions detailed in clause 3.3 of this specification, the conductors shall be retained.

When networks are being inspected particular attention shall be paid to networks that contain or show signs of :-

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- Where stays on poles have relaxed or contain un-stayed angles supports
- Where networks have been subjected to the long term installation of temp shrouding.
- Where networks have been the subject of temp repairs or diversions.

All of the above situations are likely to result in poor conductor regulation with the increased potential for conductor clashing and reduced operational performance. Where poor conductor regulation is identified the network conductors shall be unbound and re-sagged back to design tensions

Where additional phase conductors are required to convert a split phase network to a three phase network, the additional conductors shall be erected to match the existing network conductors.

The following list details allowable conductor types that may be retained on renovated networks or may be added as additional conductors

- a) 16, 32, 70 or 100mm² HD PVC covered conductor to BS6485, Type 8 (black)
- b) 16, 32, 70 or 100mm² HDBC conductor, to BS7884

Note

PVC insulated conductors to BS6485 type 8 are approved for the following situations

- Protection of BT and Network Rail circuits.
- Proximity protection to buildings, i.e. these conductors meet the requirements of the term "effectively insulated conductors" as detailed in NSP/004/011. Hence they are suitable for networks located less than 3m from a building but greater than 1m.

3.6.3 Clearances & Conductor design ratings

The minimum allowable clearances to ground and other objects on renovated LV networks and the service spans attached to those networks shall be as detailed in the overhead line clearance document NSP/004/011. Additional guidance on service clearances can also be found in NSP/004/043

Particular attention is required with regard to the location of existing networks in proximity to buildings and trees as new ESQCR legislation requires the use of insulated conductors or permanent shrouding where minimum clearances cannot be achieved.

All lines shall provide a minimum design clearance of 50°C in line with the original design criteria.

3.6.4 Configuration & spacing of conductors

3.6.4.1 Service Lines & Distribution Lines

- a) The nominal arrangement of conductors for a three-phase, four-wire service line or distribution line where a switch wire is not required shall be as follows:

Old Phase Colour	New Harmonised Colours for phases	Position on pole relative to pole top
Red	Brown	No 1
Yellow	Black	No 2
Blue	Grey	No 3
Neutral	Neutral	No 4

Where a switch wire is required, this shall occupy position No. 4 and the neutral conductor placed in the fifth position below the switch wire.

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b) Conductor spacing shall be 300mm.

c) Continuous earth Wires (CEW)

Where a CEW is required it shall be mounted on the side of the pole directly opposite the neutral position, Where the network does not already include a PL switchwire, then as an alternative the CEW may be located in the 5th or lowest position on the pole.

CEW shall be not less than 16mm² CSA. Earth wires shall be bare except where passing over Network Rail circuits when they shall be insulated.

When positioning conductors, space shall no longer be left for the addition of future additional phase conductors, as reinforcement shall be with ABC.

d) Street Lighting Switch Wire

The use of public lighting switch wires shall be discouraged, but where required they shall be 16 mm² HDBC, insulated conductors.

e) Guard wires

Guard wires of 16 mm² HDBC shall be erected above the line conductors.

3.6.5 Conductor Erection sags & tensions

Bare and insulated open wire conductors shall be erected in accordance with the sag charts shown on drawing no's 1091403404, 3405, 3506 & 3507. See appendix 2 – 5 for details. Additional guidance information on conductor erection techniques can be found in NSP/004/105.

Bare Neutral conductors shall be regulated to match in with the PVC covered conductors.

New conductors shall be pre-stressed before final make off for as long a period as is practicable but preferably not less than 20 minutes. Copper conductors shall be pre-stressed to the instructions given on the appropriate sag table.

Each appendix includes one or more table's dependant upon the requirement for the conductors to be erected at full or reduced tensions. The reduced tension tables shall be used where a section of the network is constructed with open wire conductors and is terminated onto steel poles.

Note:-

Appendixes 6 – 9 have been included to provide sag/tension information on historical plain aluminium conductors that require adjustments. ie where work other than renovation is taking place.

3.6.5.1 Service Lines

a) The PVC-covered phase conductors shall always be positioned below the neutral conductor on open wire services. See NSP/004/043 for more details on services

b) Conductor spacing shall be 150mm.

3.6.6 Services

3.6.6.1 Services –Hybrid or copper concentric

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No changes required to this type of service provided they comply with NSP/004/043 and NSP/004/011

3.6.6.2 Services - Open Wire Construction

Existing Copper bare or PVC covered service aerials may be retained provided they comply with NSP/004/043 and NSP/004/011. All neutral service connections must consist of double split bolt connectors or be replaced with compression connectors.

All hessian covered service aerials or bare/covered aluminium service aerials shall be replaced.

3.6.7 Supports

Intermediate and section support sizes (see ENA TS 43-30 Tables 2 -4) are related to loadings imposed on PVC covered conductors, the loading point being considered to be 600mm below the pole top. Stayed supports are designed in accordance with crippling loads imposed by PVC covered conductor loadings (See ENA TS 43-30 Charts 8 -10). The strength of supports are derived from information contained in BS 1990 with respect to red fir poles. The minimum design factor of safety for supports is 2.5 under the design conditions stated in clause 3.1.1 of this specification

3.6.7.1 Wood Poles

Existing wood poles may be retained provided they are of minimum classification Medium to BS1990 and that they have been examined and tested in accordance with NSP/004/112. Retained poles shall be capable of providing continued service for a minimum of 10 years.

All supports that are retained and are in excess of 30 years old shall be treated with BORON ROD preservative.

Where poles have been inspected and confirmed to be suitable for continued use but do not have 3m gouge marks to indicate the correct installed sinking depth they may be retained providing they are situated in locations that can be safely accessed using mobile access platforms or where the pole can be temporarily stayed in four directions. Where this is not the case the pole must be replaced

All replacement wood poles shall be supplied in accordance with NPS/001/001.

The replacement poles shall be minimum grade Medium as detailed in BS 1990. Poles shall be treated and fabricated in accordance with ENA TS 43-88 and drawing no. 1091193301, sht 3.

Care shall be taken when selecting replacement poles to take account of the increased pole sinking depths and the potential reduction in ground clearances. Further details on the required sinking depths can be found in NSP/004/041 clause 3.4

3.6.7.2 Steel Poles

For information on retaining or replacing existing steel poles see clause 3.5.4

3.6.8 Insulators and binders

The condition of all existing insulators and binders shall be inspected and where they are found to be in good condition they shall be retained.

Any replacement insulators shall be supplied in accordance with NPS/001/006.

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Reel Insulators shall be as detailed in drawing no. 1000439304 sht 2.
Coachscrew service insulators shall be as detailed in drawing no. 1000439305 sht 1.

3.6.9 Steelwork

All steelwork shall be supplied in accordance with NPS/001/005.

Steelwork is designed to withstand the maximum working tensions tabled in ENA TS 43-30 clause 4.1.
All steelwork shall be as specified in ENA TS 43-95.

D-Irons shall be as detailed in drawing no. 1000439512

Service Brackets shall be to the following 100043 drawing no's 1226, 1404, 9509, 9510, 9511 or 9514 as appropriate to the conductor type and service arrangement.

Any poles containing "Swan Neck" type insulator supports shall have the complete insulator/steelwork assembly replaced with D-Irons and Insulators.

Steel pole steelwork shall be to the following 100043 or 109119 drawing no's

Steel pole collars - 3508 or 3510

Steel pole neutral bars to 3505, 3506 & 3513

Steel pole mounting bar for LV Fuses 3524

3.6.10 Stays

All stays and stay assemblies shall be compliant with clause 3.5.5 of this specification and installed in accordance with NSP/004/104 or NSP/004/041. All stays except those associated with steel poles must include stay insulators.

All stays with permali inserts must be replaced with modern type 1 insulators

New or replacement stays shall utilise stay wire to 7/4.00mm grade 1150

3.6.11 Conductor Terminations and Joints

All conductor terminations and joints shall be supplied in accordance with NPS/001/002 or NPS/001/016.

See the following drawing no's for more details on

Open Wire Dead ends - 1091193155

Concentric service Dead ends - 1091193152

Intermediate / Angle Ties – 1091193156

3.6.11.1 Mid Span Joints

Mid span joints shall be avoided wherever possible, but where unavoidable, they shall be as detailed on drawing no. 1091010102. Mid span joints shall not be used over motorways or railways.

3.6.11.2 Section Fuses

All porcelain type section fuses shall be replaced with new fuse units with lugged connections to drawing no. 1091471530.

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3.6.12 Fasteners and Washers

All Fasteners and fixings shall be supplied in accordance with NPS/001/011.

3.6.13 Technical Requirements for Open Wire Construction

3.6.13.1 Maximum and recommended span lengths

Conductor or cable size	Recommended Span (m)	Maximum Span (m)
16/25/35mm ² Concentric cables	20	30
16mm ² Service Span	20	30
16mm ² Service Line	50	50
32/70/100mm ² Distribution Lines	50	60

3.6.13.2 Wind loading Spans

All wind loading spans are based on the loadings imposed by PVC covered conductors.

Allowable wind loading spans for replacement poles are based on poles being supplied with sizes and available strengths in line with BS 1990. Consequentially Medium grade poles are suitable for 2, 3, 4 or 5 wire networks with wind spans up to and including 50m.

For spanning in excess of 50m or for information on the maximum windspan capability of existing poles reference shall be made to tables 2 - 4 of ENA TS 43-30.

3.6.13.3 Staying arrangements and strut loading for angle and terminal supports

Stayed supports shall be dressed as shown on drawing no's 1091193326 or 1091193327. For further guidance on allowable staying arrangements see NSP/004/104.

Stay spreads will normally be such that an angle of 35° is provided between stay and pole. (ie stay and the vertical). As a general rule, therefore, the stay spread on horizontal ground will be equal to the length of the pole less its sinking depth. In all cases the minimum allowed stay angle shall be 30°.

Typical pole grade / minimum stay spread selection data has been provided for 4 wire Bare and PVC insulated copper networks in appendix 5 & 6. (Note the tabulated data takes no account of additional transverse loadings resulting from service aerials that are additive to the angle loadings).

Alternative conductor arrangements or staying solutions are available by manually checking the strut loads and stay wire capabilities against ENA TS 43-30 table 5 and the appropriate stay charts for the conductor size and number of wire.

Where alternative stay checks are required, users shall utilise the group 2 arrangements as defined in ENA TS 43-30 table 6.

Under exceptional circumstances where no other traditional stayed support can be utilised due to wayleave or access problems, a strut pole arrangement as shown on drawing 1091193331 may be utilised. Strut poles are limited to a maximum angle of deviation of 30°

3.6.14 Relationships with other bodies

Where lines to this standard are erected over or alongside the plant of Telecommunications Operators or Network Rail operators, then the provisions of the relevant joint agreements shall apply. Details of the

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required clearances and any special provisions are available in Company Code of Practice NSP/004/011.

Additional information with respect to Telecommunication requirements can be found in EB/BT2 and PO5/2.

The requirements of Waterways Authorities may also be found in Company Code of Practise NSP/004/011.

3.6.15 Safety Signs, labels and notices

All poles shall be fitted with safety signs and pole numbers which are compliant with NPS/001/011 and guidance document NSP/004/109

3.6.16 Earthing

See NSP/004/041 clause 3.11 for more detail on the appropriate earthing policies.

3.6.17 Connections

Connections for open wire networks shall be used according to the material of the conductors, and typical designs are shown on drawings referred to in NSP/004/107.

Before any connection is made, the conductor shall be thoroughly cleaned in order to remove surface oxidation.

Care must be taken to ensure that connectors are properly tightened and due regard paid in the case of split bolt connectors to conductor relaxation and bedding down of strands. All neutral connections shall consist of double split bolt connectors or compression fittings.

At section positions the tail end of the conductors shall be short on the supply side and long on the remote side to avoid handling a long live tail when making disconnections with the network alive.

3.6.18 Services

Services spans and connections shall be carried out in accordance with NSP/004/043

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4. References

4.1 External Documentation

Reference	Title
43-12	Insulated Aerial Bundled Conductor Erection Requirements for LV Overhead Distribution Systems
43-13	Aerial Bundled Conductors Insulated with Cross Linked Polyethylene for low voltage overhead lines.
43-14	Conductor Fittings and Associated Apparatus for use with Low Voltage Insulated Aerial Bare Conductors.
43-30	Low Voltage Overhead Lines On Wood Poles
43-88	Treatment of Wood Poles and associated Timber for overhead Lines
43-91	Stay Strands & Stay Fittings for Overhead Lines
43-92	Stay Strands & Stay Fittings for Overhead Lines
43-95	Steelwork for Overhead Lines
BS 1990	Wood Poles for Overhead Power and Telecommunications Lines part 1
EB/BT2	Conditions for BT and electricity suppliers - Joint use of poles
ESQCR	The Electricity Safety, Quality and continuity Regulations 2002, SI 2665
PO5/1	Protection of Telecommunication lines from Power Lines

4.2 Internal documentation

Reference	Title
NSP/004/011	Guidance on Overhead Line Clearances
NSP/004/041	Specification for LV ABC Overhead Lines
NSP/004/043	Specification for Overhead Services, Surface Wiring and ABC Eves Mains
NSP/004/104	Guidance notes on Stay types, make-off's and spreads
NSP/004/105	Guidance notes on conductor erection and dismantling
NSP/004/106	Guidance notes on conductor joints, terminations & binders
NSP/004/107	Guidance notes on conductor jumpers & non-tension connections
NSP/004/109	Guidance notes on anti climbing devices, signs and notices plates
IMP/010/011	Code of practise for earthing LV Networks and HV Distribution Substations
NPS/001/001	Technical Specification for Wood Poles and Associated Products for Overhead Lines
NPS/001/002	Technical Specification for Helical Products
NPS/001/005	Technical Specification for Steelwork, Conductor/Insulator Fittings and Stay Fittings for Overhead Lines and General Construction Works
NPS/001/007	Technical Specification for Conductors for Overhead Lines
NPS/001/010	Technical Specification for Notice Plates and Signs for Use on NEDL and YEDL Distribution Systems
NPS/001/011	Technical Specification for Fasteners And Fixings For Wood Pole Overhead Lines and General Construction Works
NPS/001/013	Technical Specification for Stay wire for Overhead Lines
NPS/001/016	Technical Specification for Compression Fittings and Mechanical Conductor Fittings for Overhead Lines
NPS/001/020	Technical Specification for Ground Anchor Systems for Use in Overhead Lines

4.3 Summary of Changes

Clause	Changes or amendments
Note	This document has been significantly re-arranged and clauses re-numbered, the following clauses have been identified as being of significance to the re-issue of this document

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2.2	Table of contents added
3.2	This clause defines the objectives and requirements for renovating networks
3.3	This clause defines extensive maintenance and the minimum requirements that must be achieved to allow an existing open wire network configurations to be retained
3.4	This clause confirms typical how networks shall be refurbished when the requirements of clause 3.3 cannot be complied with and the network is required to be converted to ABC
3.5.3	The requirement for all existing poles without gouge marks to be replaced has been removed. A clearer statement has been added confirming the requirement that all new poles are treated with non creosote based preservatives and a reminder has been inserted about changes in sinking depths when selecting replacement poles
3.5.4	Additional detail included within this clause to highlight how to test a steel pole and the requirement to re-appraise the need for steel poles as they are an expensive line structure
3.6.1	A statement has been added allowing the selective use of permanent shrouding as an alternative to converting a network to ABC.
3.6.2	A new clause has been added requiring open wire networks to be re-sagged where the conductors are out of regulation.
3.6.3	A new clause has been added confirming the need to ensure that when a network is refurbished this also includes a requirement to ensure that all service spans comply with statutory ground clearances.
3.6.8	A new clause has been added requiring all old swan neck type insulator arrangements to be replaced.
Appendix 12	New clause added to provide guidance on identifying existing pole grades
Appendix 13	New clause added to provide additional guidance on ESQCR refurbishment schemes

5. Definitions

Term	Definition
PME	Protective Multiple Earthing
PILC	Paper Insulated Lead Covered Cable
MEN	Multiple Earthed Neutral
Anchor Clamp	A fitting, which transfers the tensile forces within the tensioned conductor, cores to the supporting structure.
Suspension Clamp	A fitting that encloses and supports all ABC cores at an intermediate support.
Insulation Piercing Connectors (IPC's)	Connectors that can be applied to insulated conductors without the prior removal of the insulation
Main Lines	Lines of 4 core ABC with main cores of 50, 70, 95 or 120mm ² conductor
Service Lines	Lines of 2 or 4 core ABC with core sizes of 35mm ² conductor
Service Span	The span in which the service conductors are attached to a building at reduced tensions.
Design Loading Conditions	These are the augmented ice and wind loadings corresponding to ice 9.5mm diameter, wind 380N/m ² , as specified in EATS 43-12
Terminal Support	A support where the ABC is attached by an anchor clamp and does not continue into an adjacent span.
Intermediate Support	A support where the ABC system is attached by a suspension clamp. Intermediate supports may have angles of deviation upto 60°.
Section Support	A support where adjacent spans are attached by means of anchor clamps. Section supports may have angles of deviation and may have provisions for electrically sectionalising the system. Unless there is a specific need for the conductors to be cut, the ABC shall be run unbroken and a loop, approximately 0.5m deep shall be provided.

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Standards Manager		

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
A Bilclough	Overhead Programme Manager		

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
A Bilclough	Overhead Programme Manager		

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator		

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager		

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
M.Nicholson	Head of System Strategy		

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Appendix 1 - Index of drawings for conversion of existing open wire networks to ABC construction or renovation of existing open wire arrangements

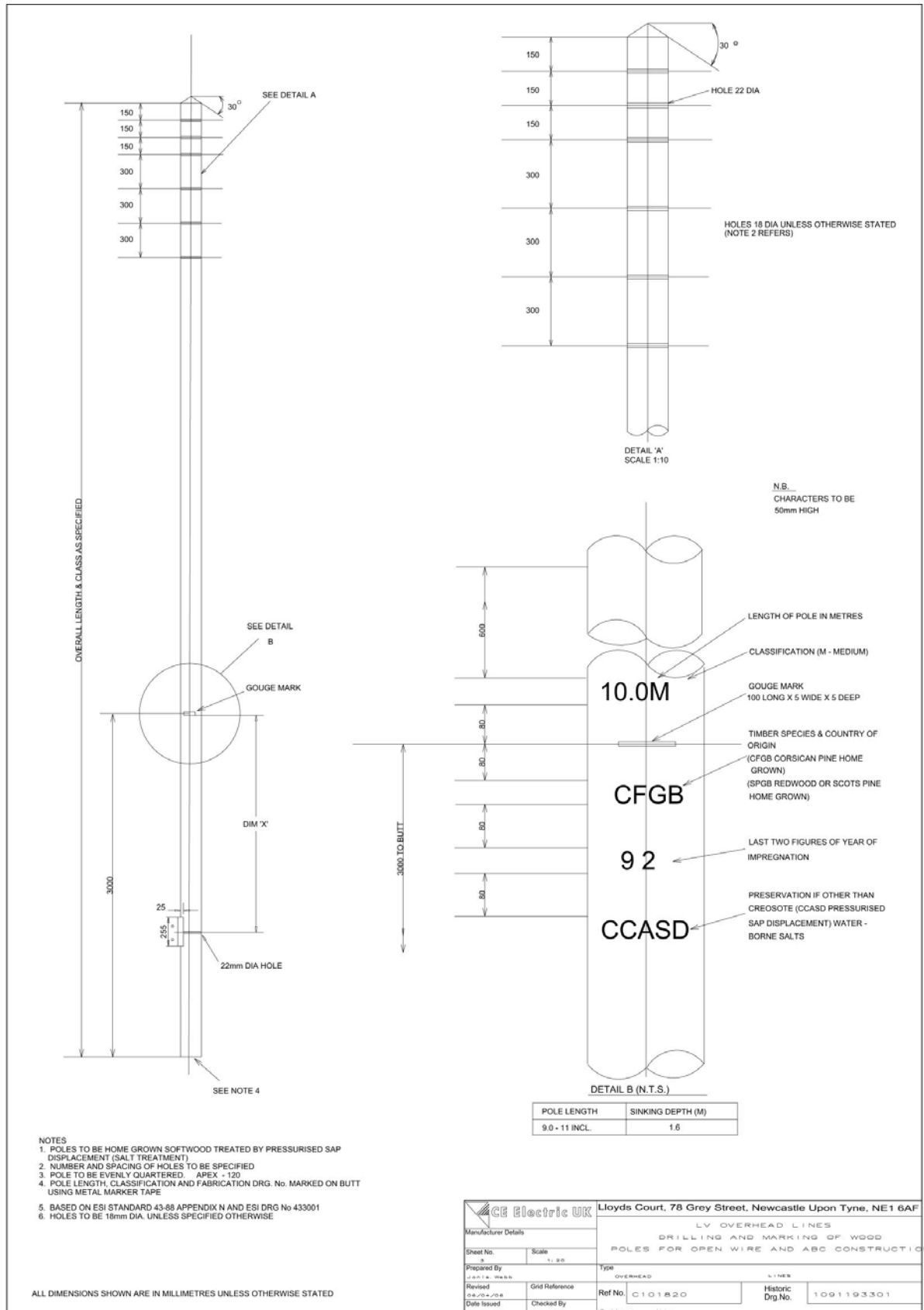
Wood Poles

Drilling and Marking of Wood Poles for Open Wire Arrangements	1091193301 Sht 3
Renovation of Open Wire Straight Line Pole to ABC Const	1091193302
Renovation of Open Wire Angle Pole (30° dev) to ABC	1091193303 Sht 1
Renovation of Open Wire Angle Pole (60° dev) to ABC	1091193303 Sht 2
Renovation of Open Wire Section Angle (90° dev) to ABC	1091193304
Renovation of Intermediate Angle Pole with Tee off to ABC	1091193305
Renovation of Intermediate Angle Support to Typical ABC Tee off	1091193306
Renovation of Open Wire Terminal Pole to ABC	1091193307
Renovation of Open Wire Terminal Pole to ABC (C/w Cable Box)	1091193308
Renovation of Open Wire Int. Pole with Offset bracket to ABC	1091193309
Arrangement of 7/4.00mm stay assembly on LV wood Pole	1091193326

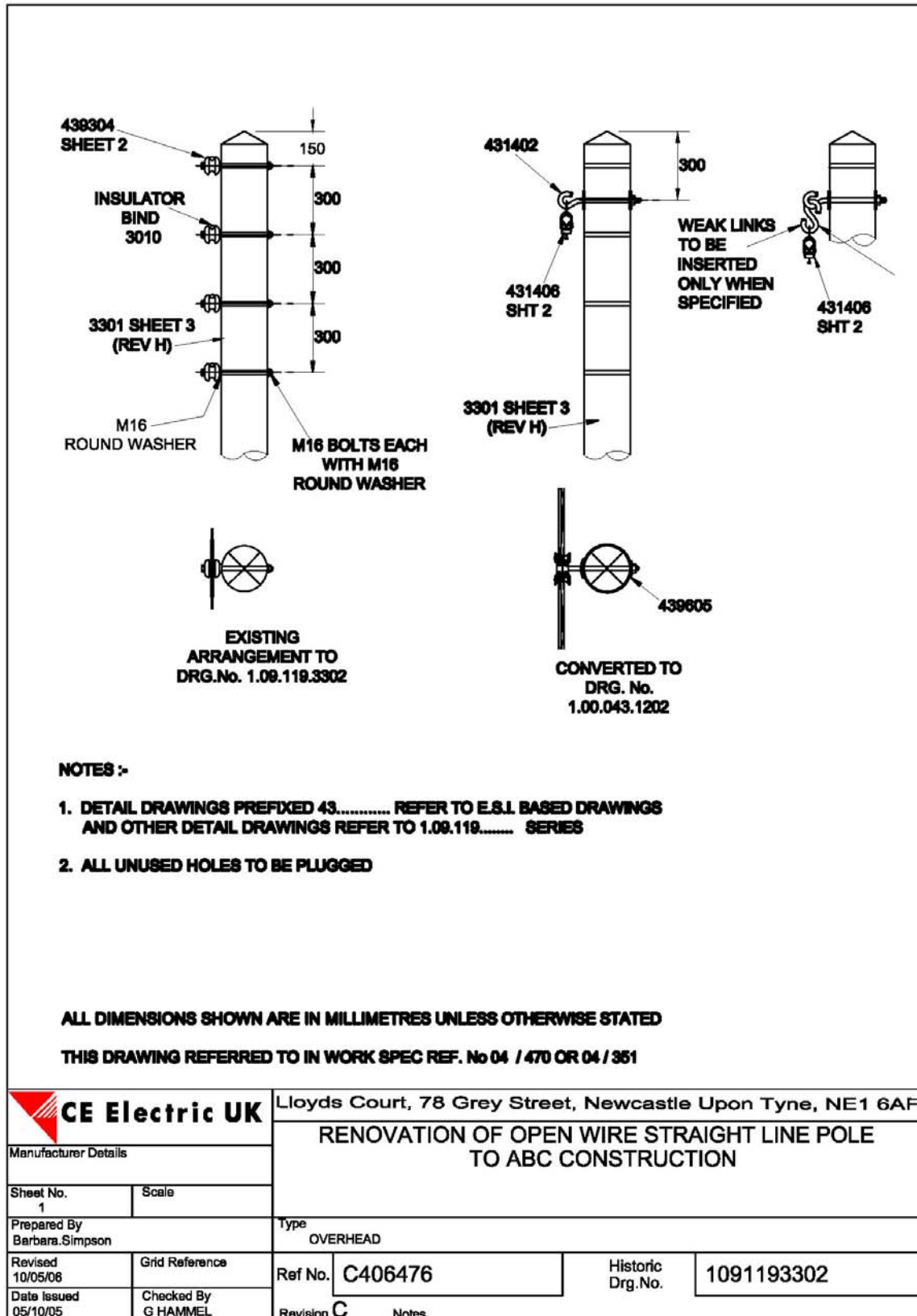
Steel Poles

Steel Pole Details	1091193501 Sht 2
Foundation Requirements for Steel Poles	1091193502 Sht 1
Protective Coating for Base of LV Steel Poles	1091193502 Sht 2
ABC Collars for Steel Poles	1091193510 Sht 2
Cable Cleats for Steel Poles	1091193514 Sht 4
Bonding Connection on J & K Steel Poles	1091193525
Intermediate Support	1091193601 Sht 2
Intermediate Support up to 30° Deviation	1091193603 Sht 2
Section Pole - Straight Line or Angle	1091193605 Sht 1
Terminal Support	1091193606
Typical Service Arrangement on Steel Poles	1091193609
Earthing Connection for Steel Poles	1091193622
Conversion of steel Pole Tops on Undrilled Poles	1091193661
ABC or open wire networks on predrilled Steel Poles	1091193662

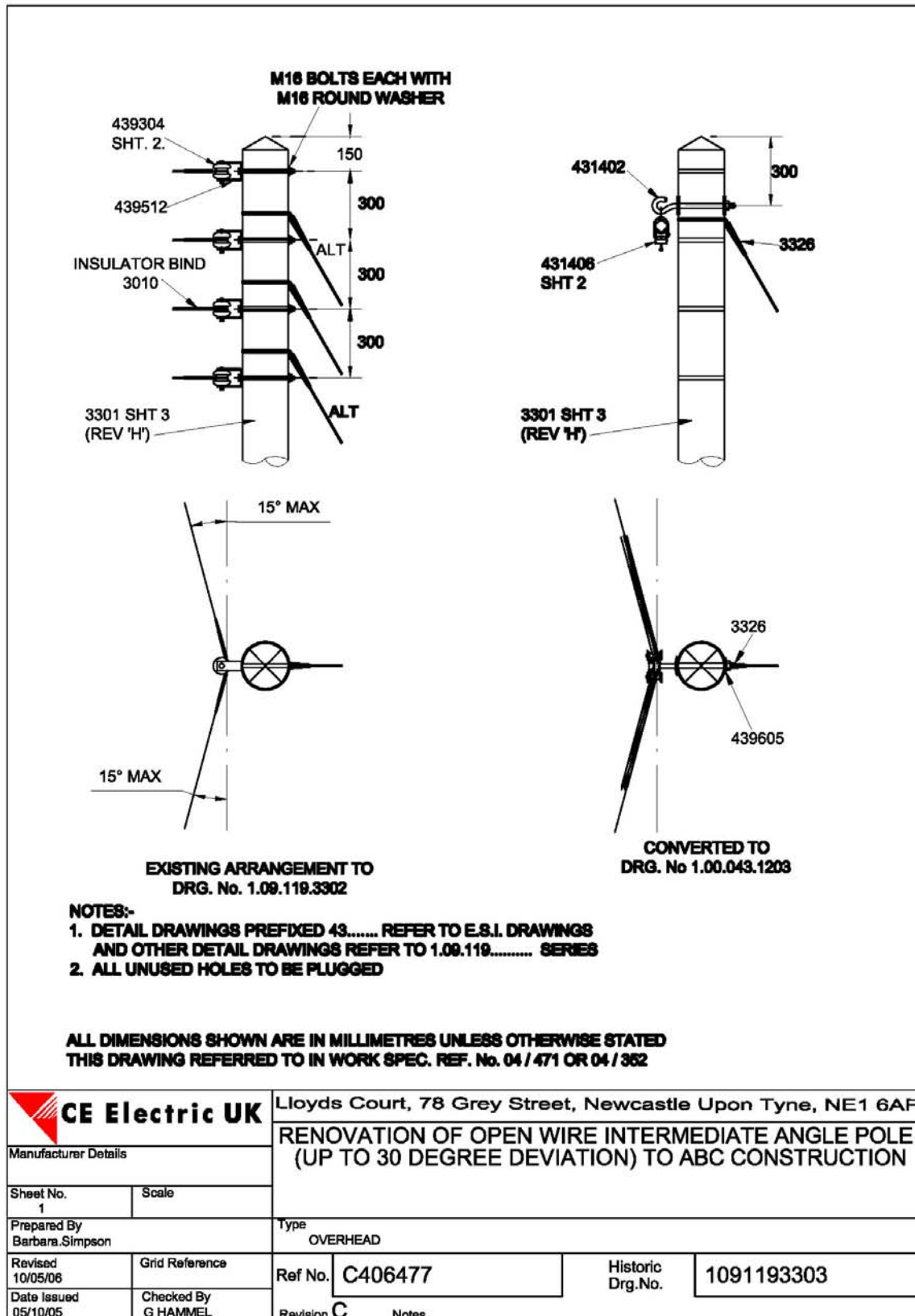
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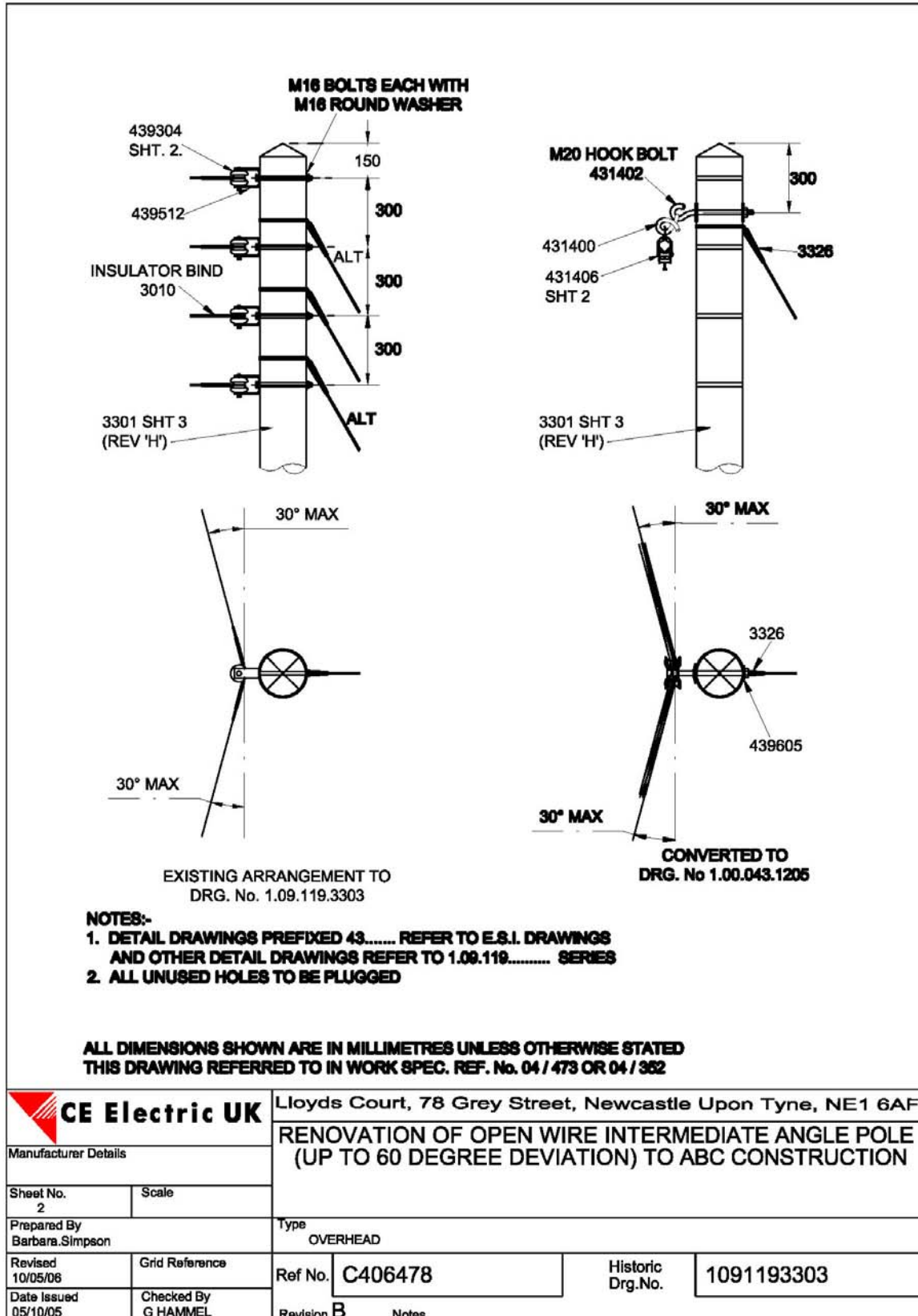
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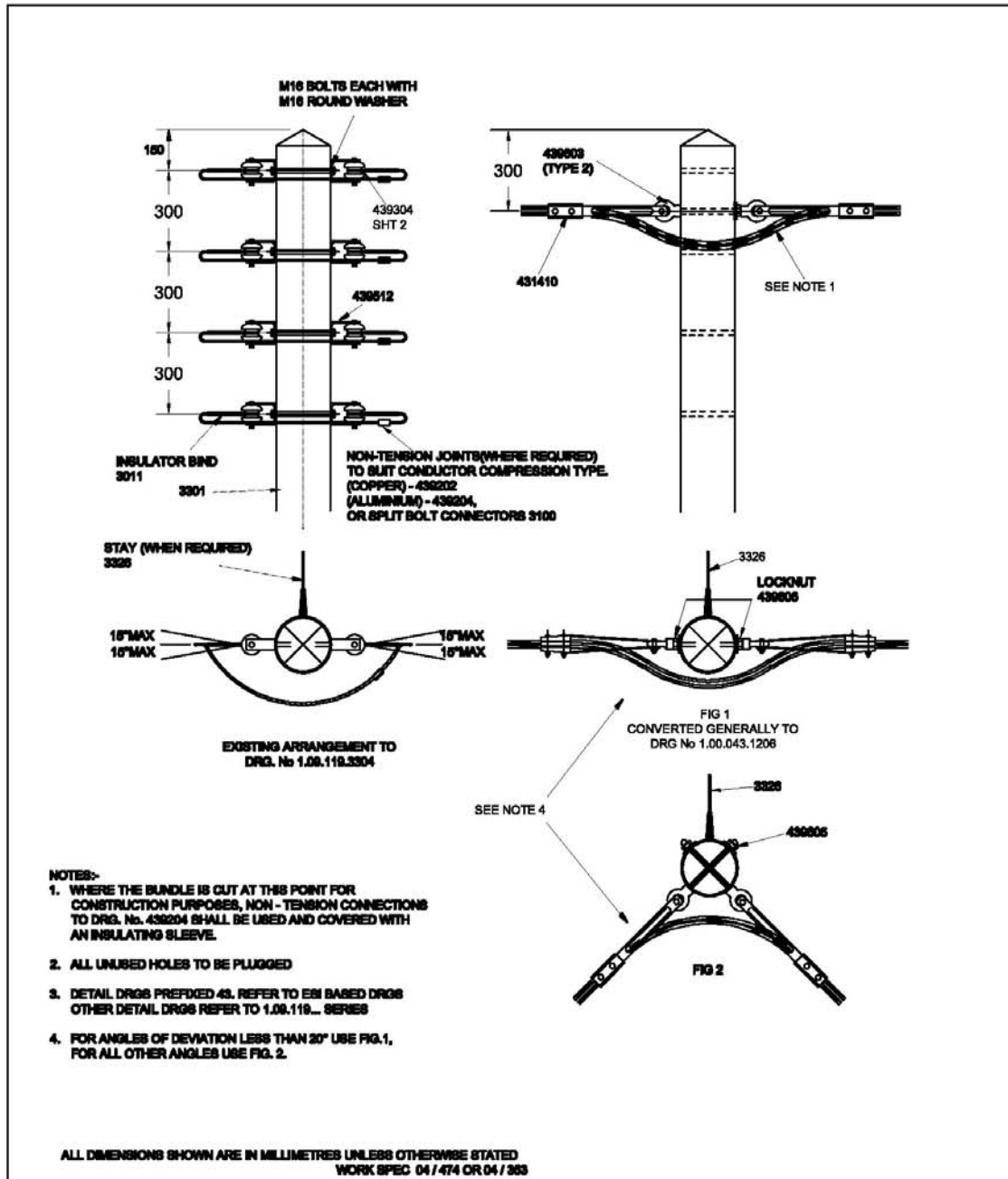
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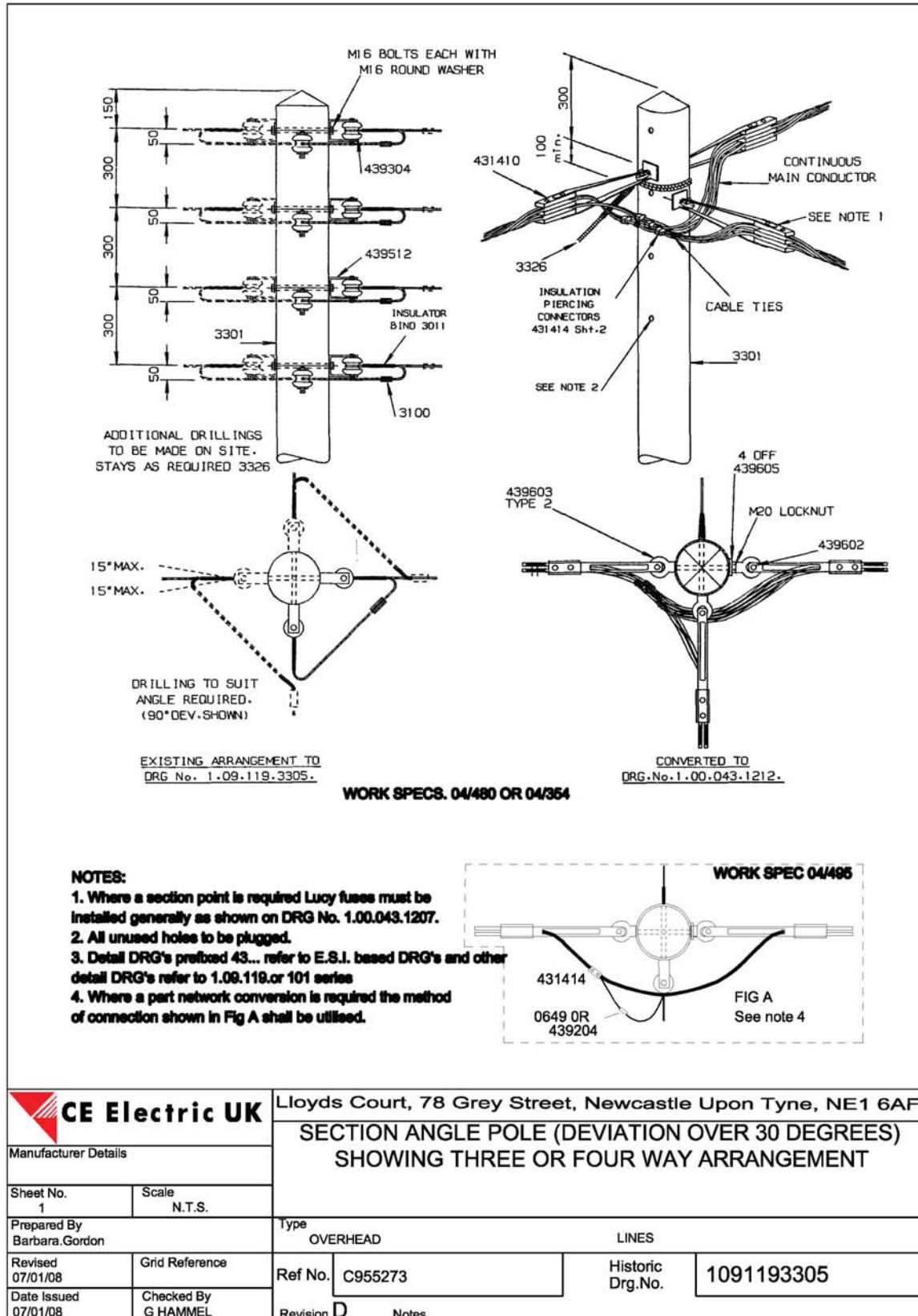


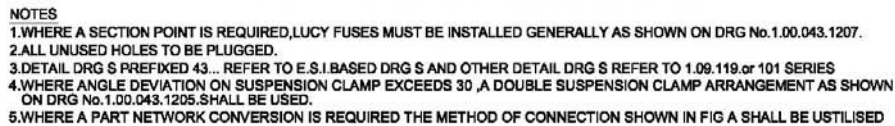
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CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		RENOVATION OF OPEN WIRE SECTION ANGLE POLE (DEVIATION UP TO 90 DEGREES) TO ABC SECTION SUPPORT	
Sheet No. 1	Scale		
Prepared By Barbara Simpson		Type OVERHEAD	
Revised 10/05/06	Grid Reference	Ref No. C101825	Historic Drg.No. 1091193304
Date Issued	Checked By G HAMMEL	Revision A Notes	

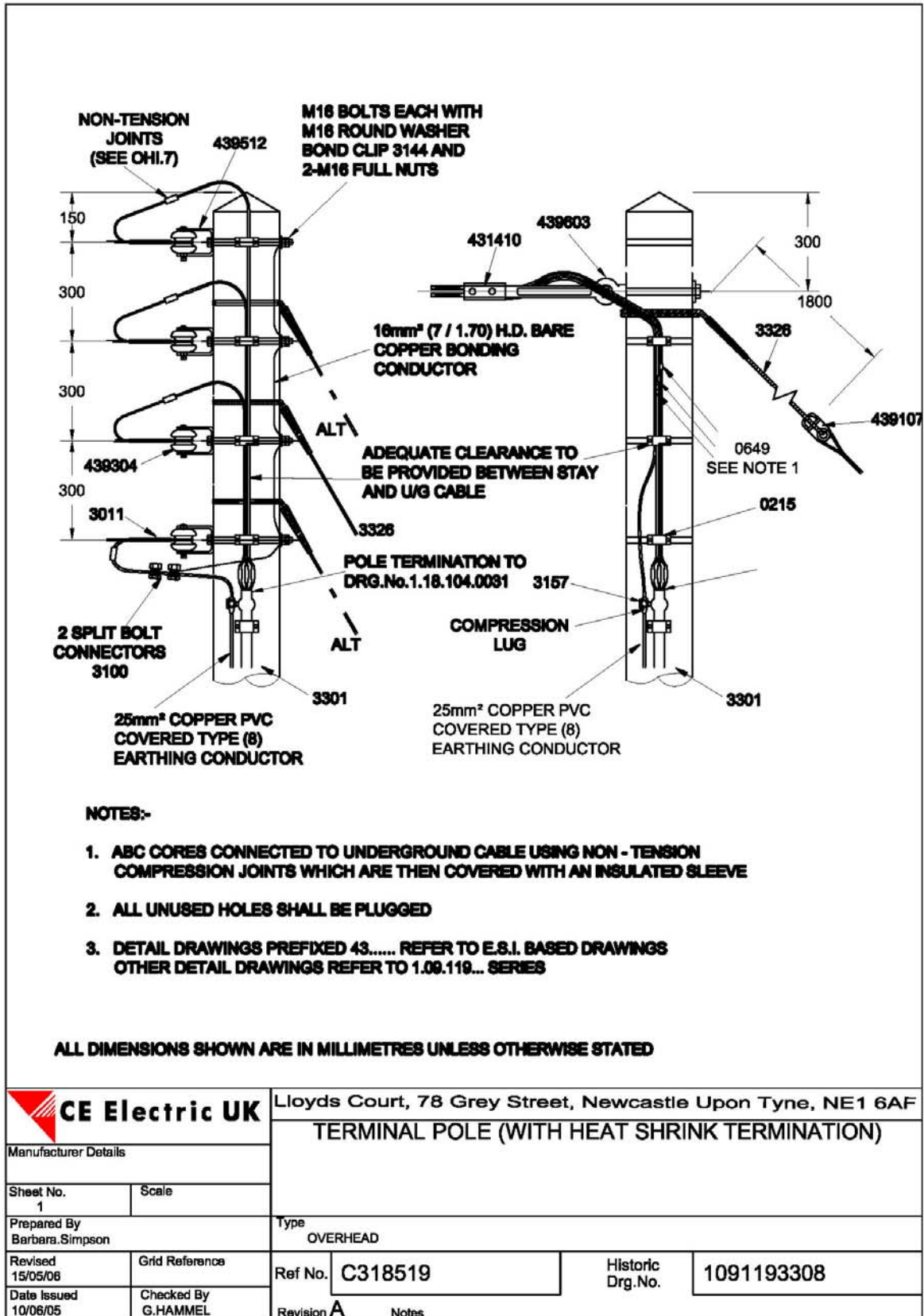
Document reference		NSP/004 /041/001		
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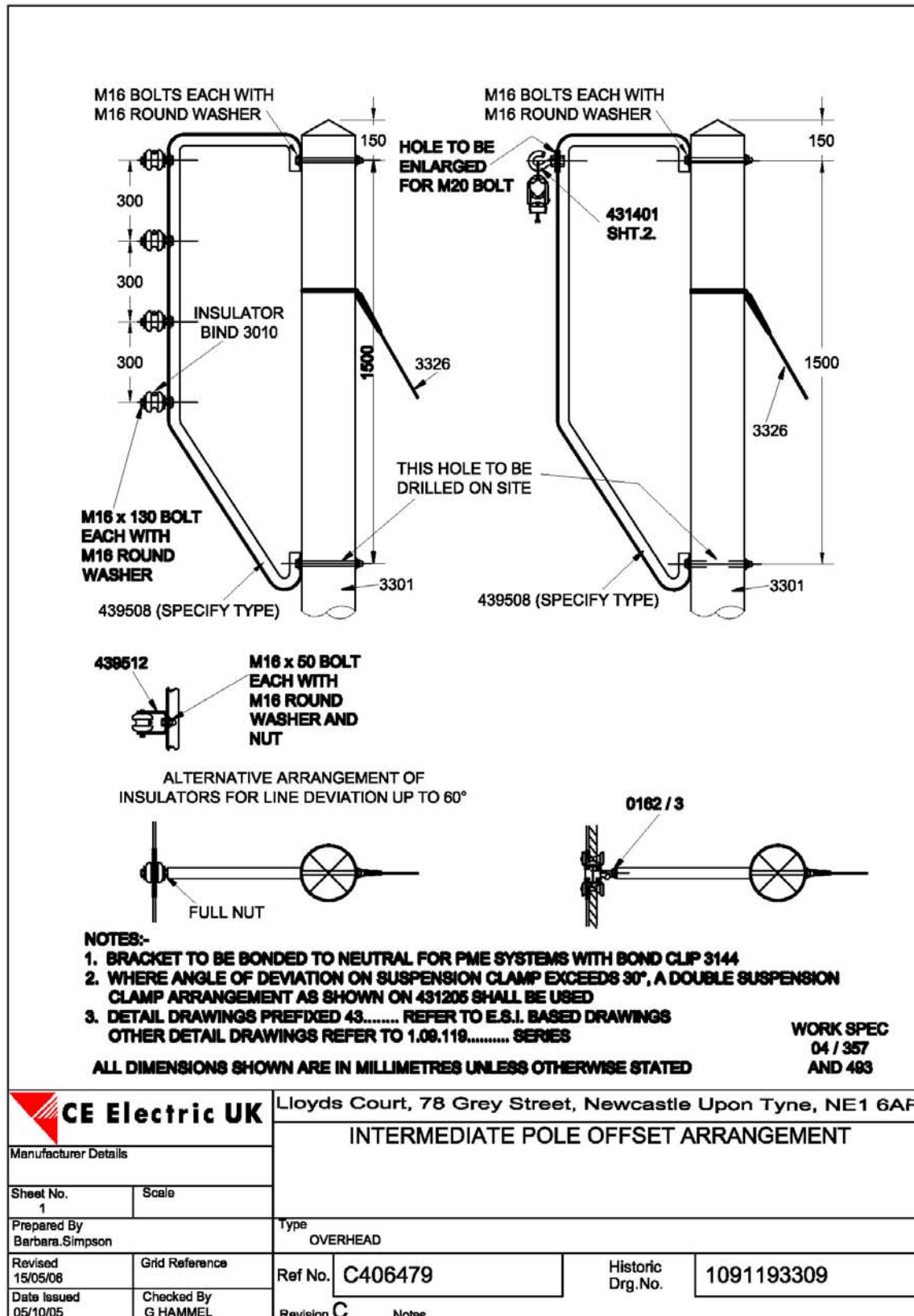


 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		INTERMEDIATE ANGLE POLE WITH TEE-OFF CONNECTION		
Sheet No. 1	Scale	Type OVERHEAD		
Prepared By Barbara Simpson		Ref No. C316835		
Revised 15/05/06	Grid Reference	Historic Drg.No.		1091193306
Date Issued 11/01/01	Checked By G. HAMMEL	Revision C Notes		

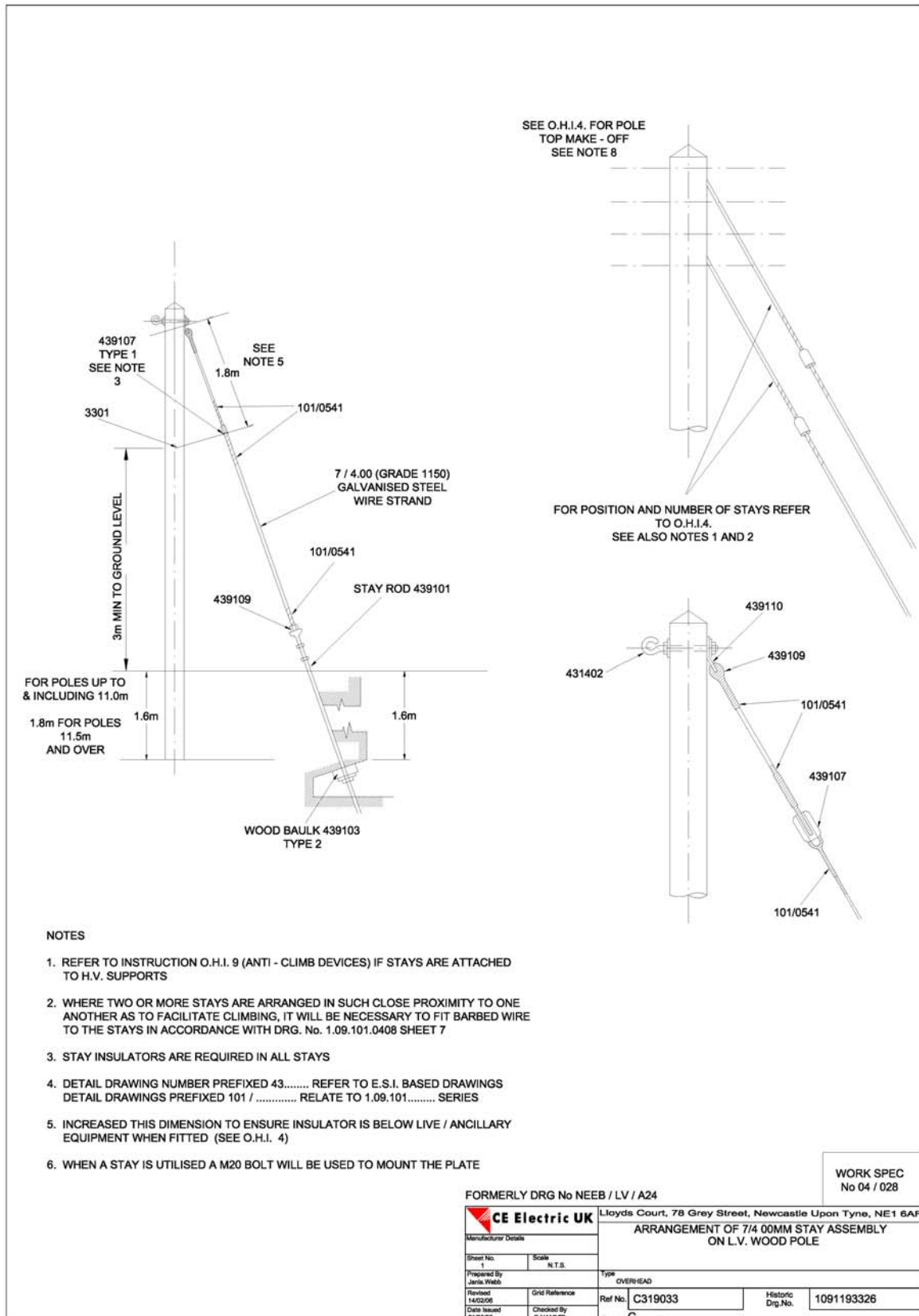
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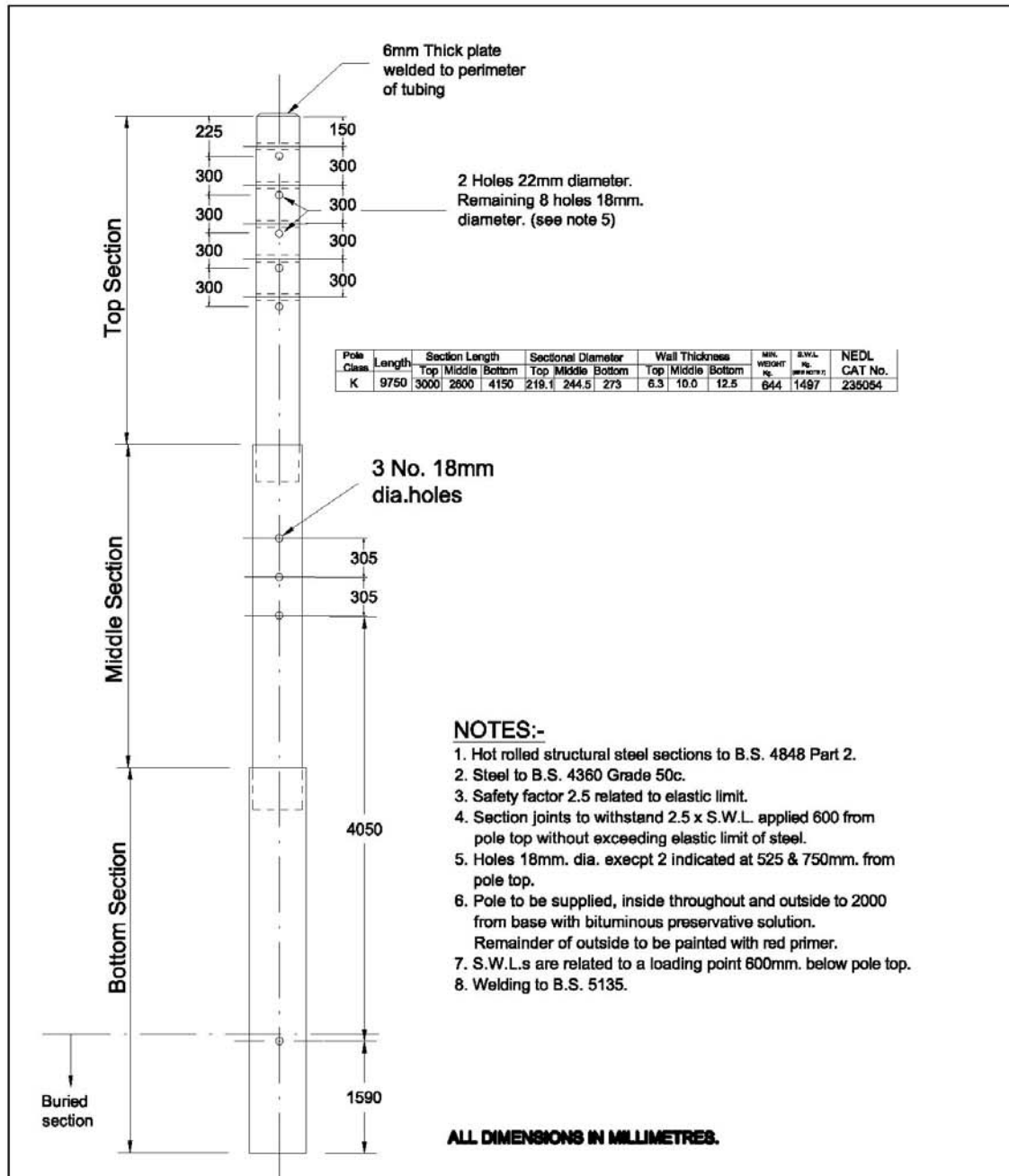
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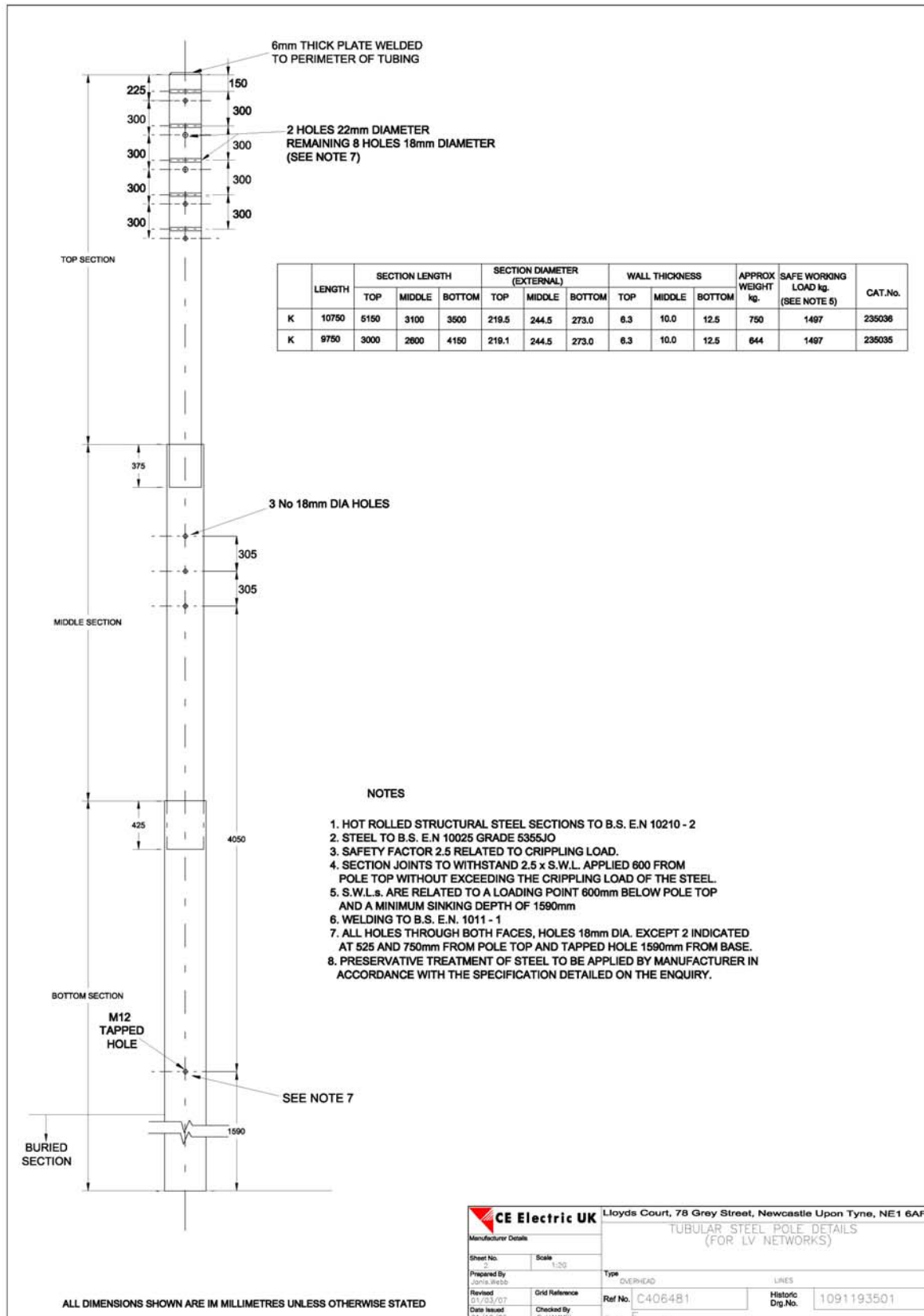
Steel Poles

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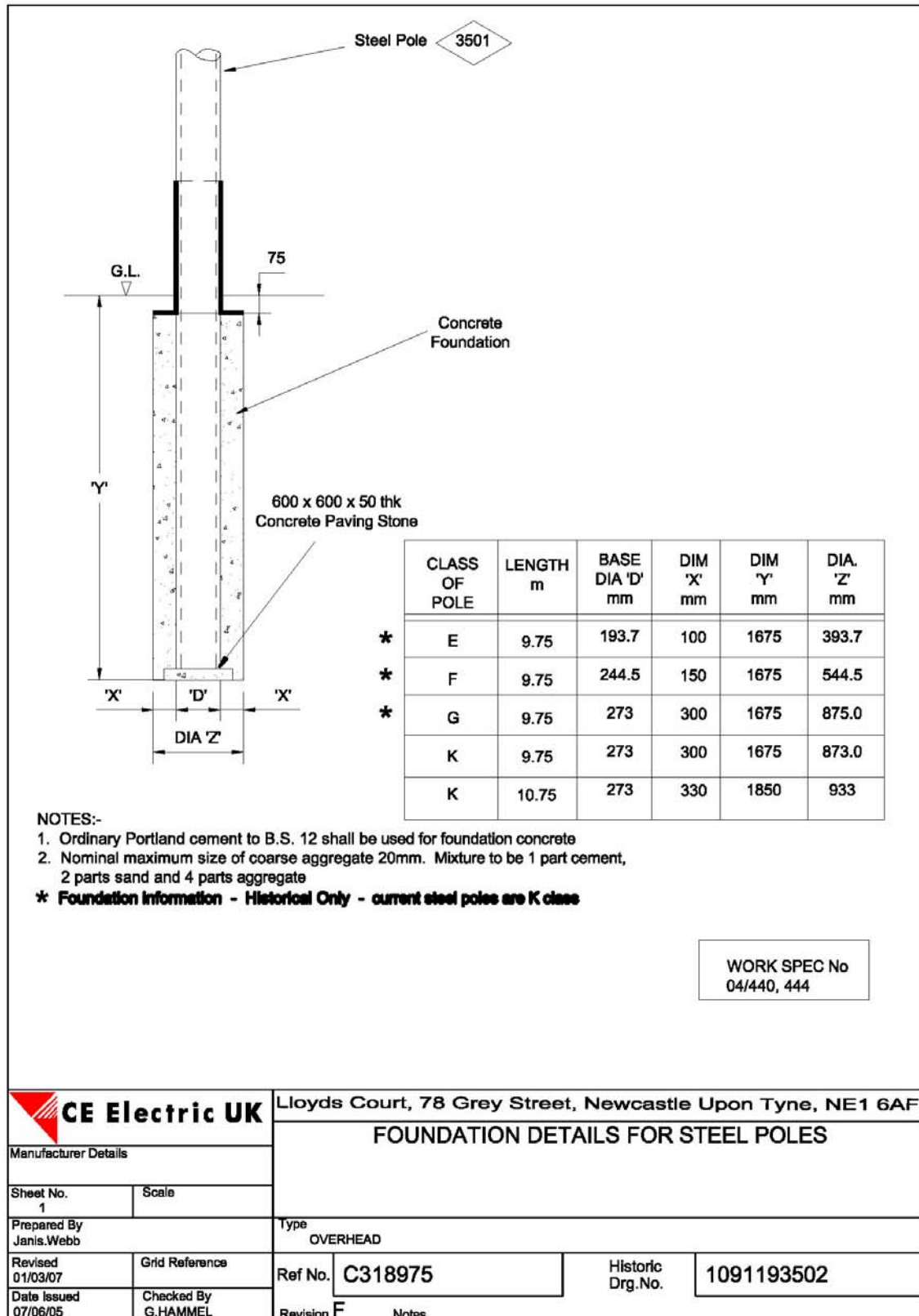


CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		STEEL POLE DETAILS		
Sheet No. 1	Scale N.T.S.			
Prepared By Jonis.Webb		Type OVERHEAD	LINES	
Revised 28/02/07	Grid Reference	Ref No. C406480	Historic Drg.No.	1091193501
Date Issued 30/09/05	Checked By G HAMMEL	Revision F	Notes	

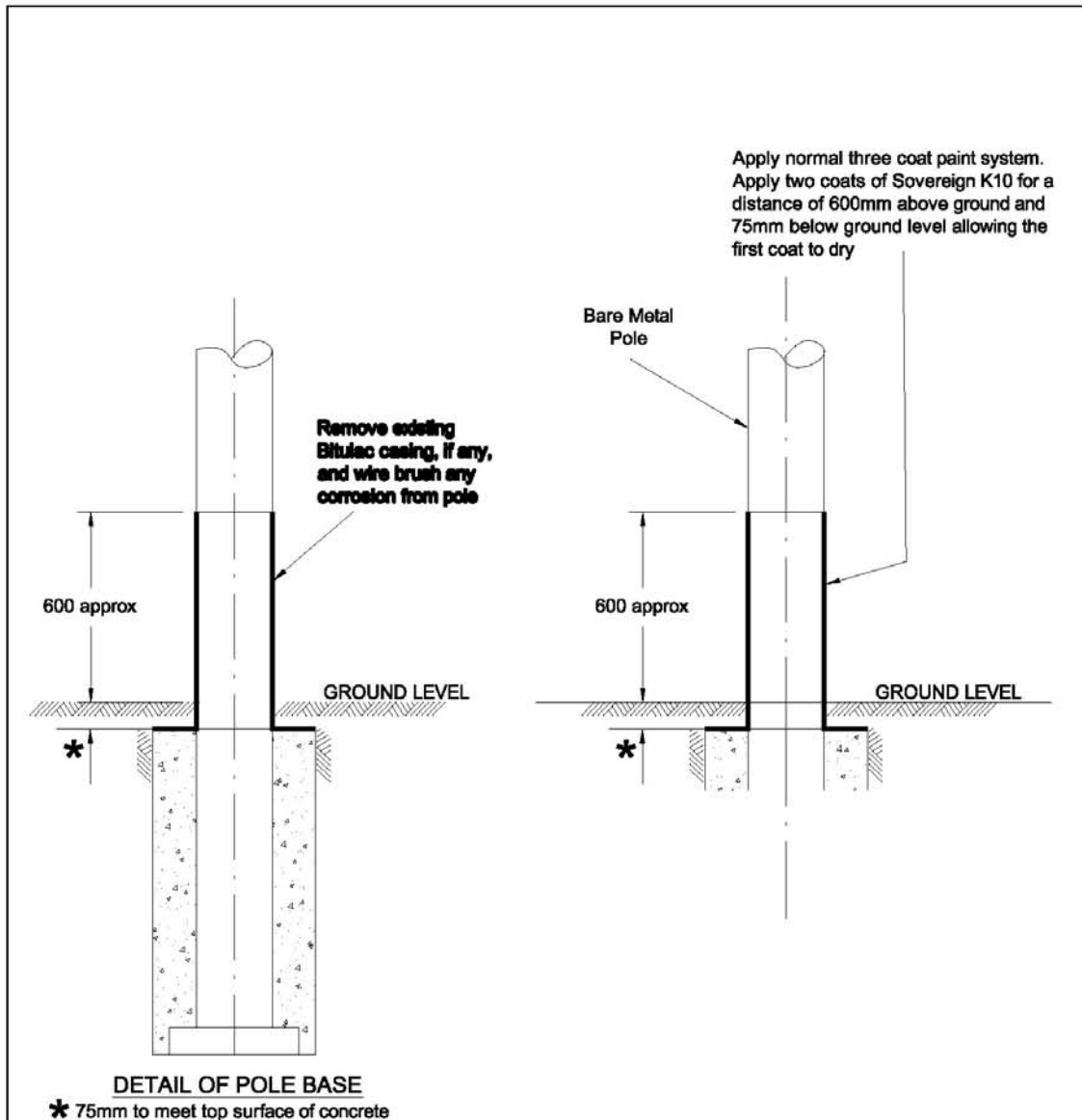
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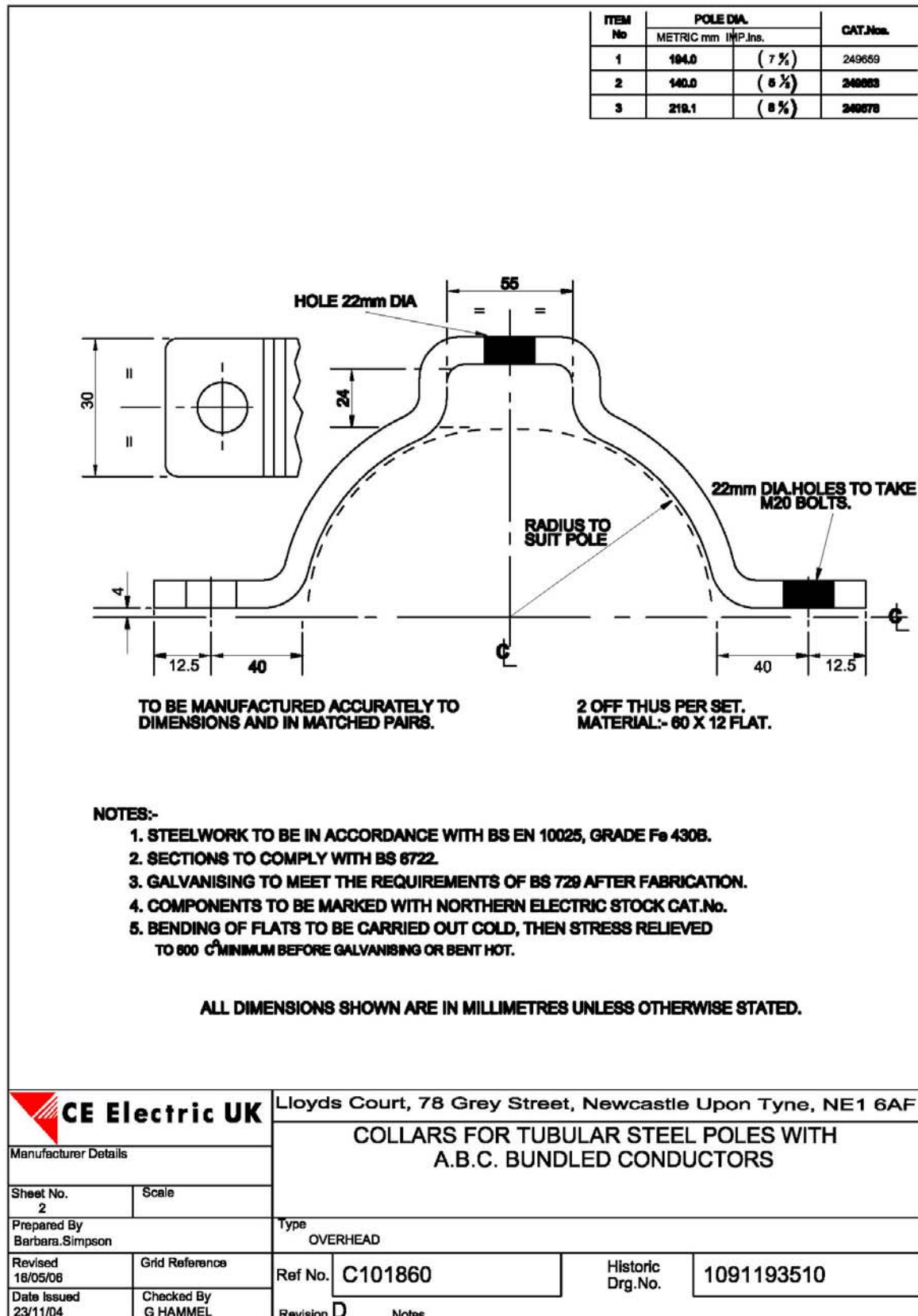
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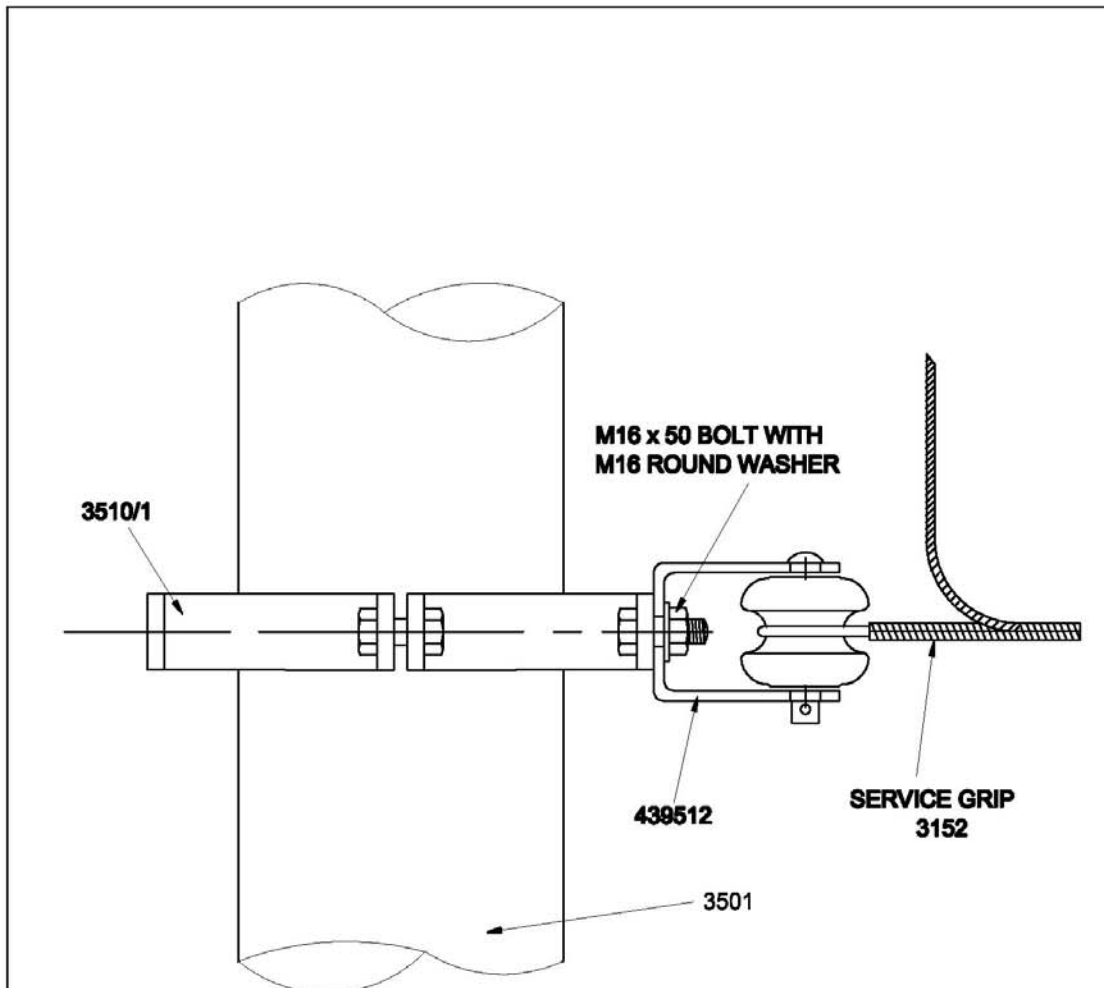
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All dimensions shown are in millimetres unless otherwise stated

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Manufacturer Details		PROTECTIVE COATING FOR BASE OF LV STEEL POLE			
Sheet No. 2	Scale	Type OVERHEAD			
Prepared By Barbara Simpson	Grid Reference	Ref No.	C406482	Historic Drg.No.	1091193502
Revised 16/05/06	Checked By G HAMMEL	Revision	C	Notes	
Date Issued 30/09/05					

Document reference		NSP/004 /041/001		
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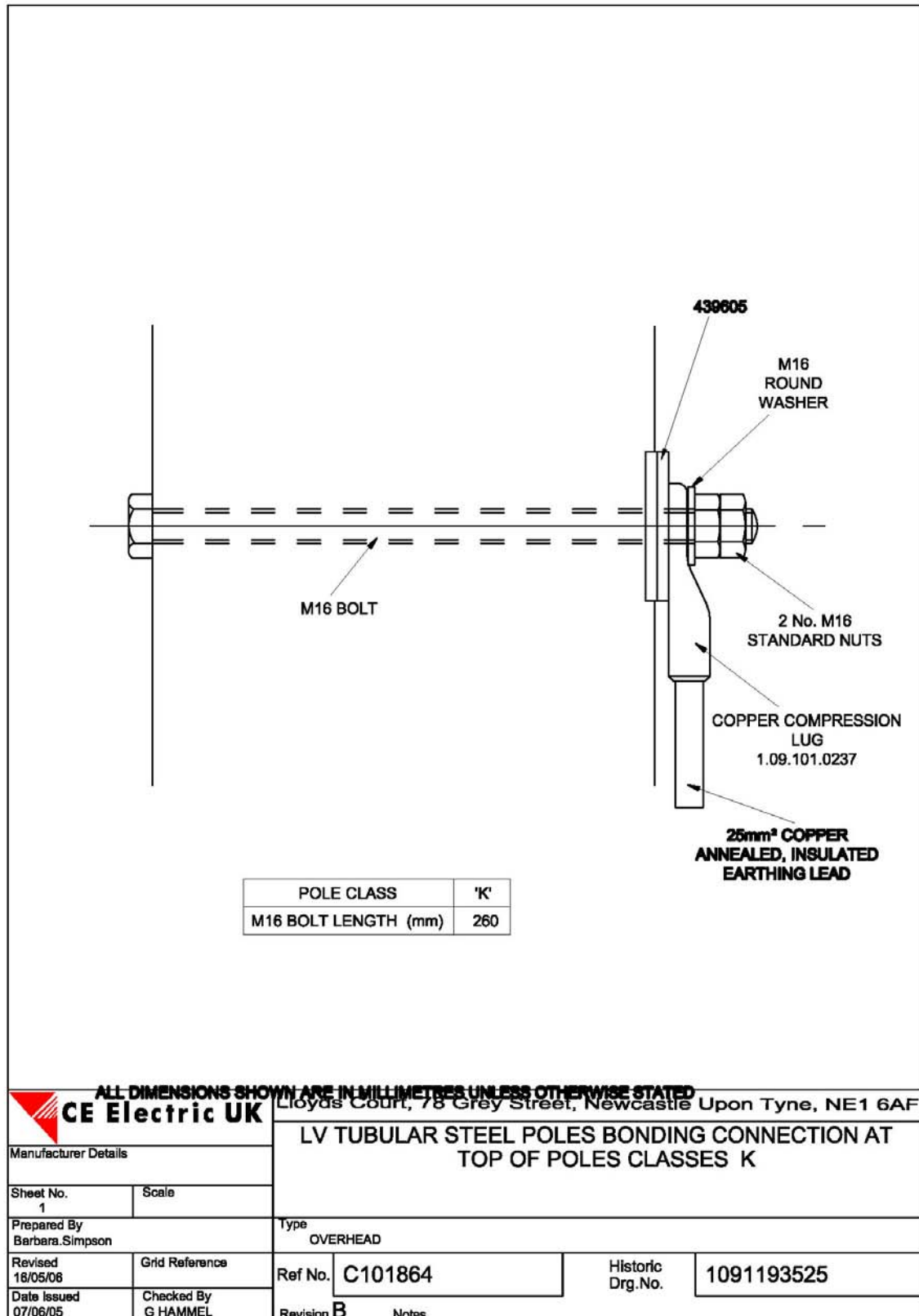


NOTES

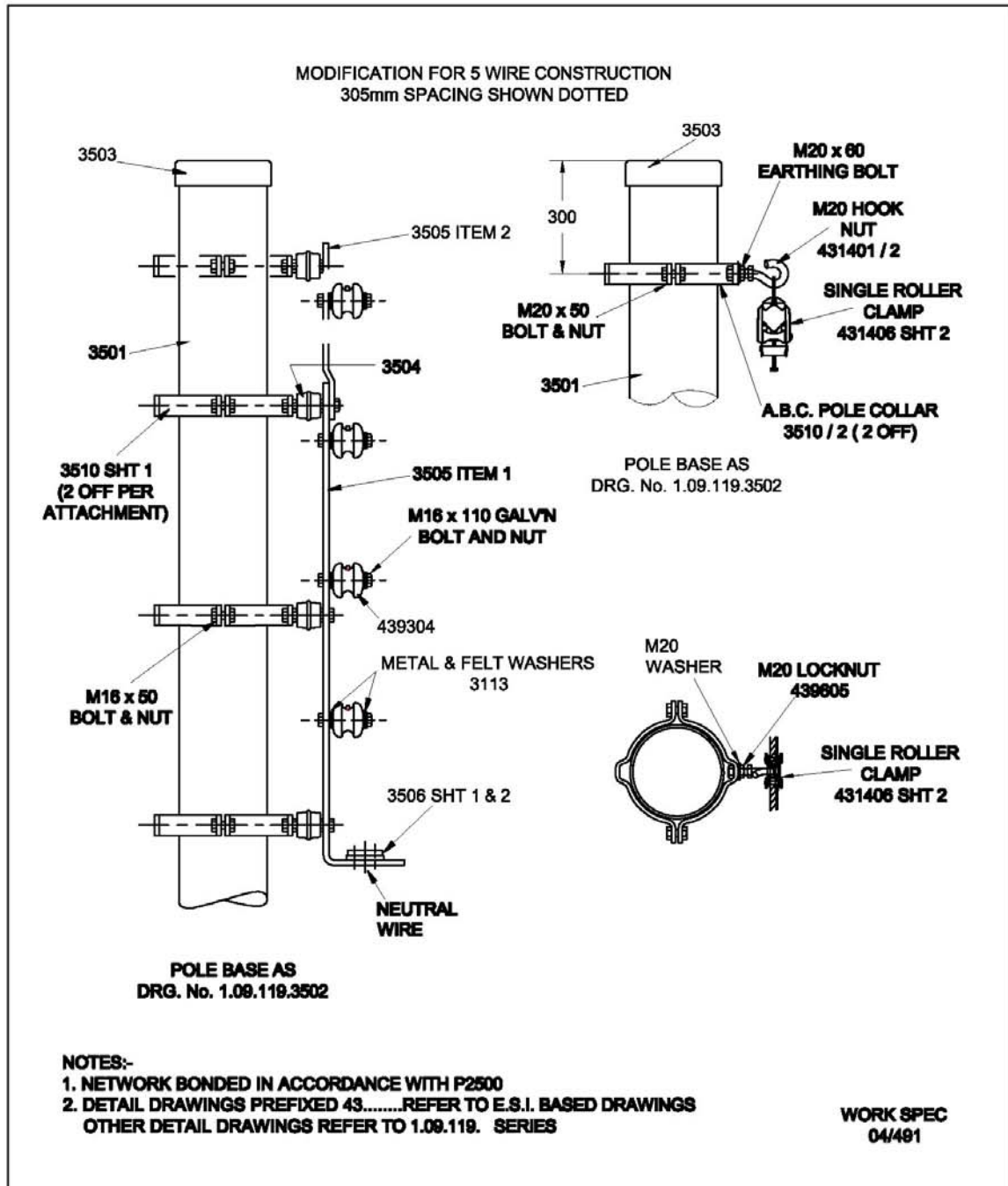
1. DETAIL DRAWINGS PREFIXED 43..... REFER TO E.S.I. BASED DRAWINGS
OTHER DETAIL DRAWINGS REFER TO 1.08.119. SERIES
2. WHERE STEEL POLES ARE OF THE PRE DRILLED TYPE, THE SERVICE D - IRON
MAY BE DIRECTLY ATTACHED USING NEAREST SPARE HOLE

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		TYPICAL SERVICE ARRANGEMENT AT STEEL POLES			
Sheet No. 1	Scale	Type OVERHEAD			
Prepared By Barbara Simpson					
Revised 18/05/08	Grid Reference	Ref No.	C406483	Historic Drg.No.	1091193509
Date Issued	Checked By G HAMMEL	Revision	A	Notes	

Document reference		NSP/004 /041/001			
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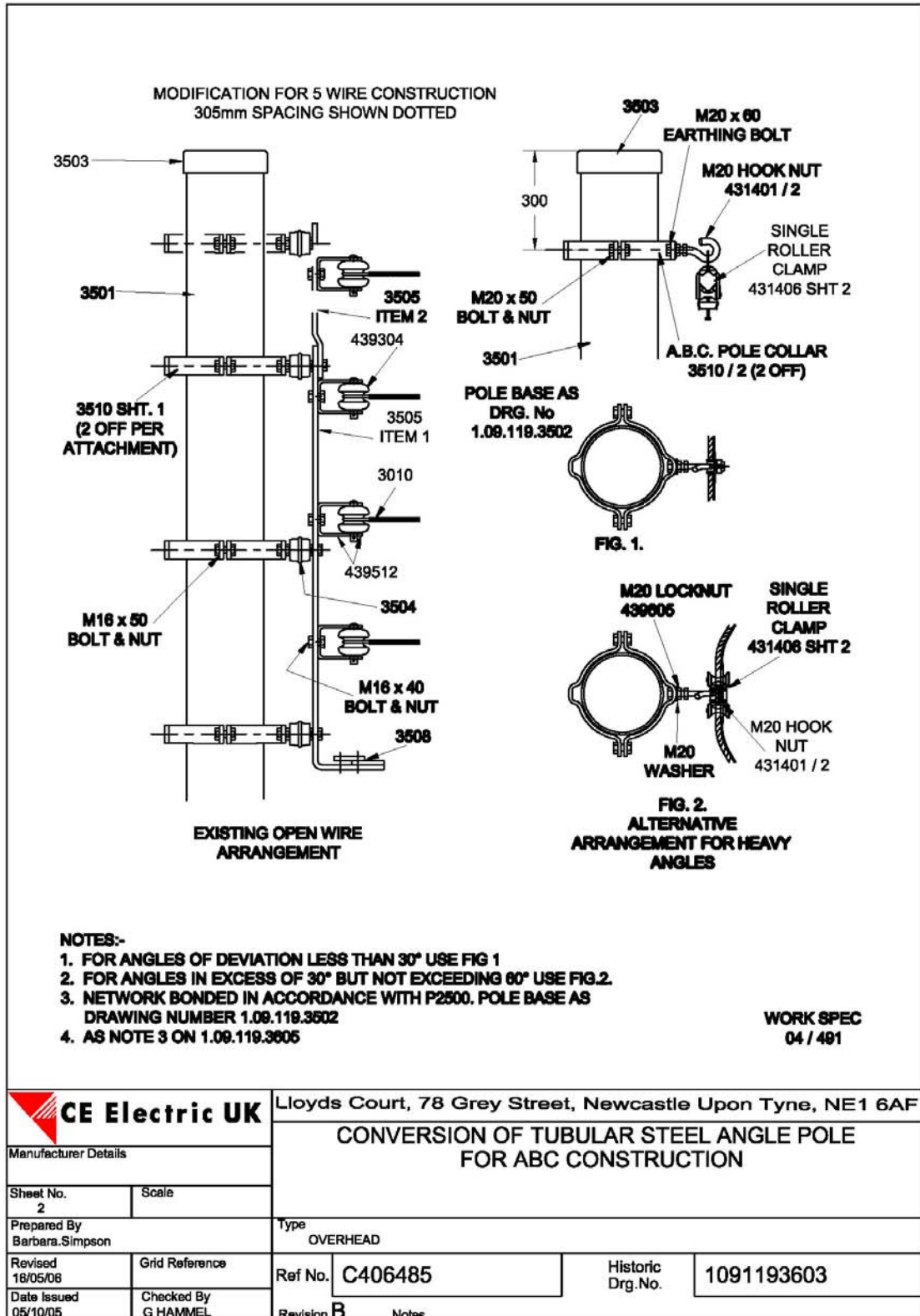


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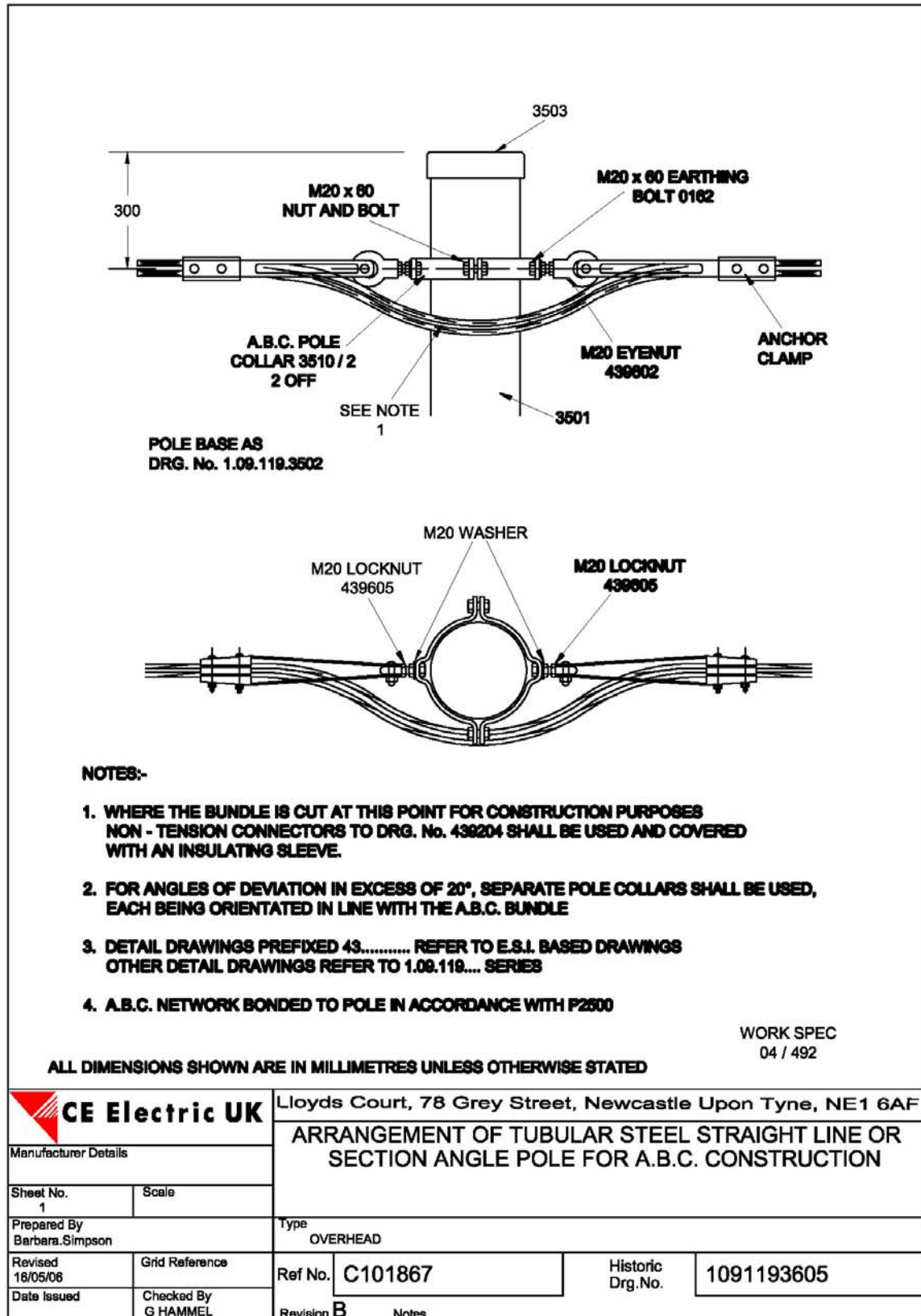


 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF		
Manufacturer Details		CONVERSION OF TUBULAR STEEL STRAIGHT LINE POLE FOR ABC CONSTRUCTION		
Sheet No. 2	Scale	Type OVERHEAD		
Prepared By Barbara.Simpson		Ref No. C406484		
Revised 16/05/06	Grid Reference	Historic Org.No.	1091193601	
Date Issued 05/10/05	Checked By G HAMMEL	Revision B	Notes	

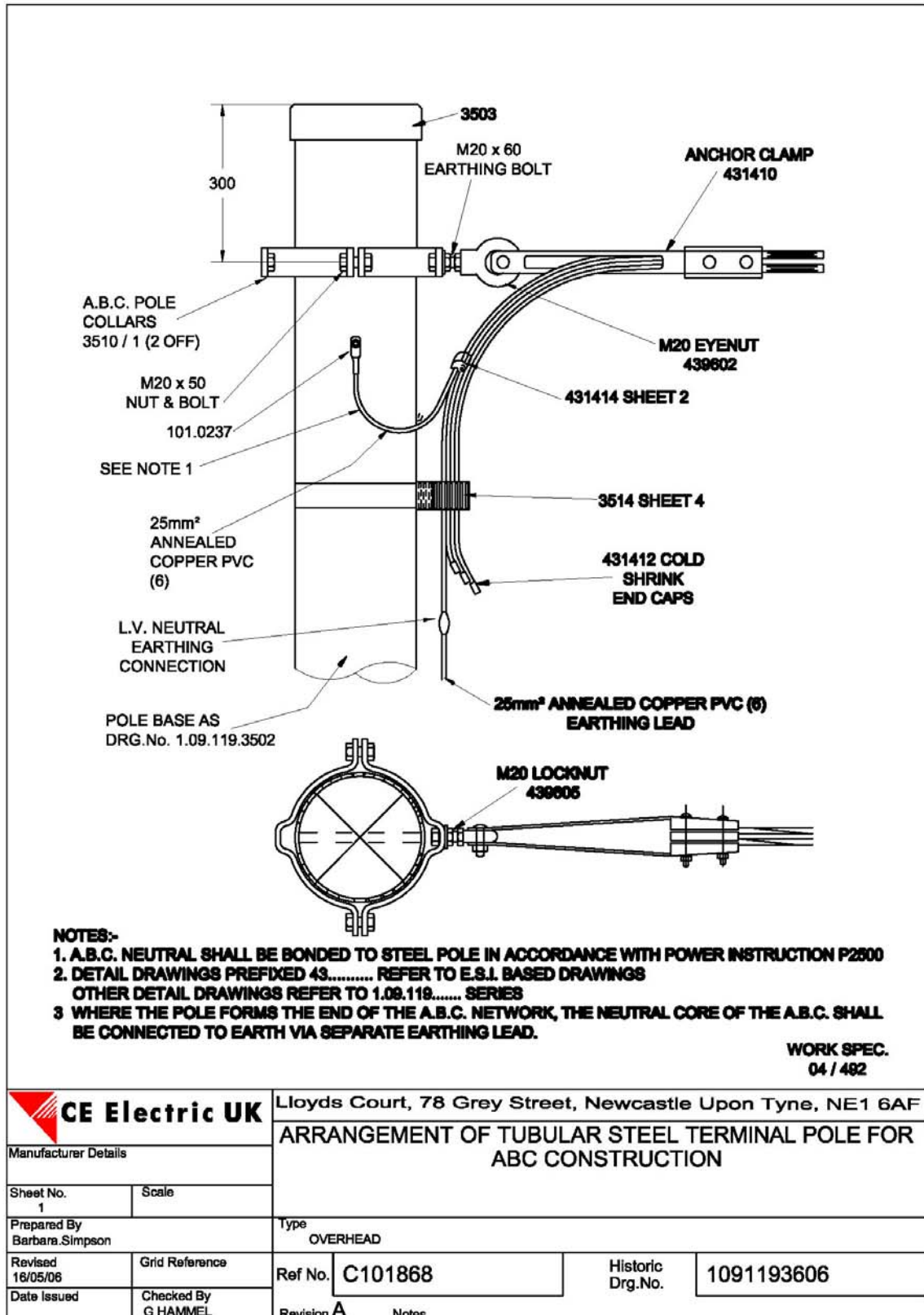
Document reference	NSP/004 /041/001				
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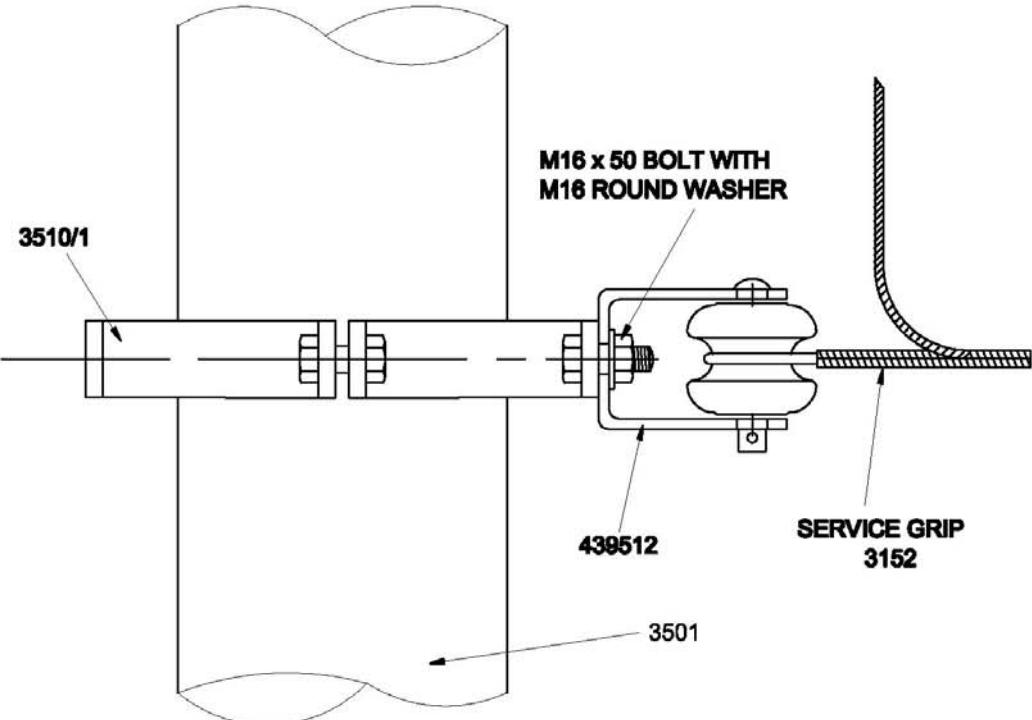
Document reference		NSP/004 /041/001			
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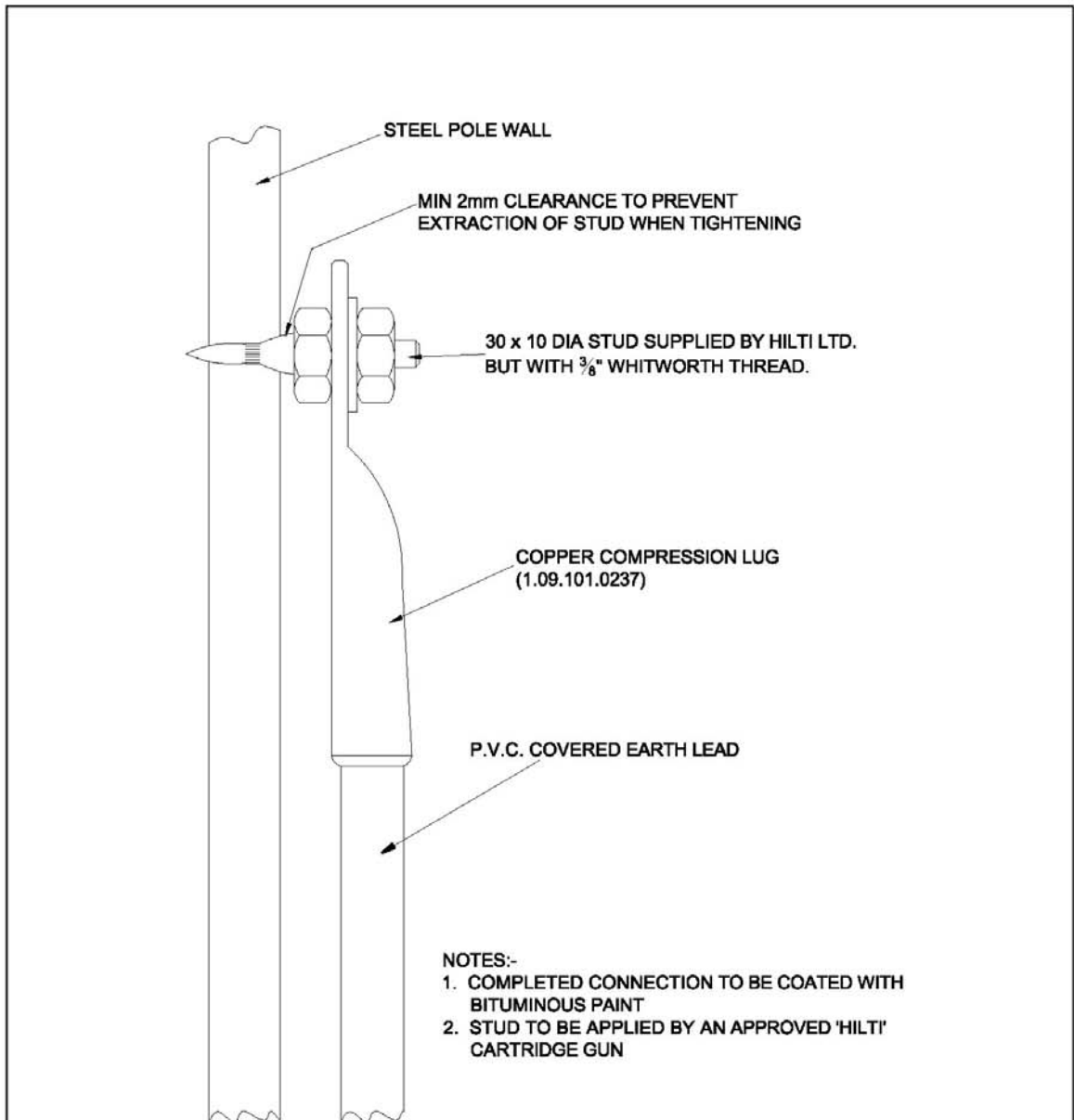
Document reference		NSP/004 /041/001			
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Document reference		NSP/004 /041/001			
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 NOTES 1. DETAIL DRAWINGS PREFIXED 43..... REFER TO E.S.I. BASED DRAWINGS OTHER DETAIL DRAWINGS REFER TO 1.08.119. SERIES 2. WHERE STEEL POLES ARE OF THE PRE DRILLED TYPE, THE SERVICE D - IRON MAY BE DIRECTLY ATTACHED USING NEAREST SPARE HOLE	
CE Electric UK	
Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
TYPICAL SERVICE ARRANGEMENT AT STEEL POLES	
Manufacturer Details	
Sheet No. 1	Scale
Prepared By Barbara Simpson	
Type OVERHEAD	
Revised 16/05/06	Grid Reference
Date Issued	Checked By G HAMMEL
Ref No.	C406483
Historic Drg.No.	1091193509
Revision	A
Notes	

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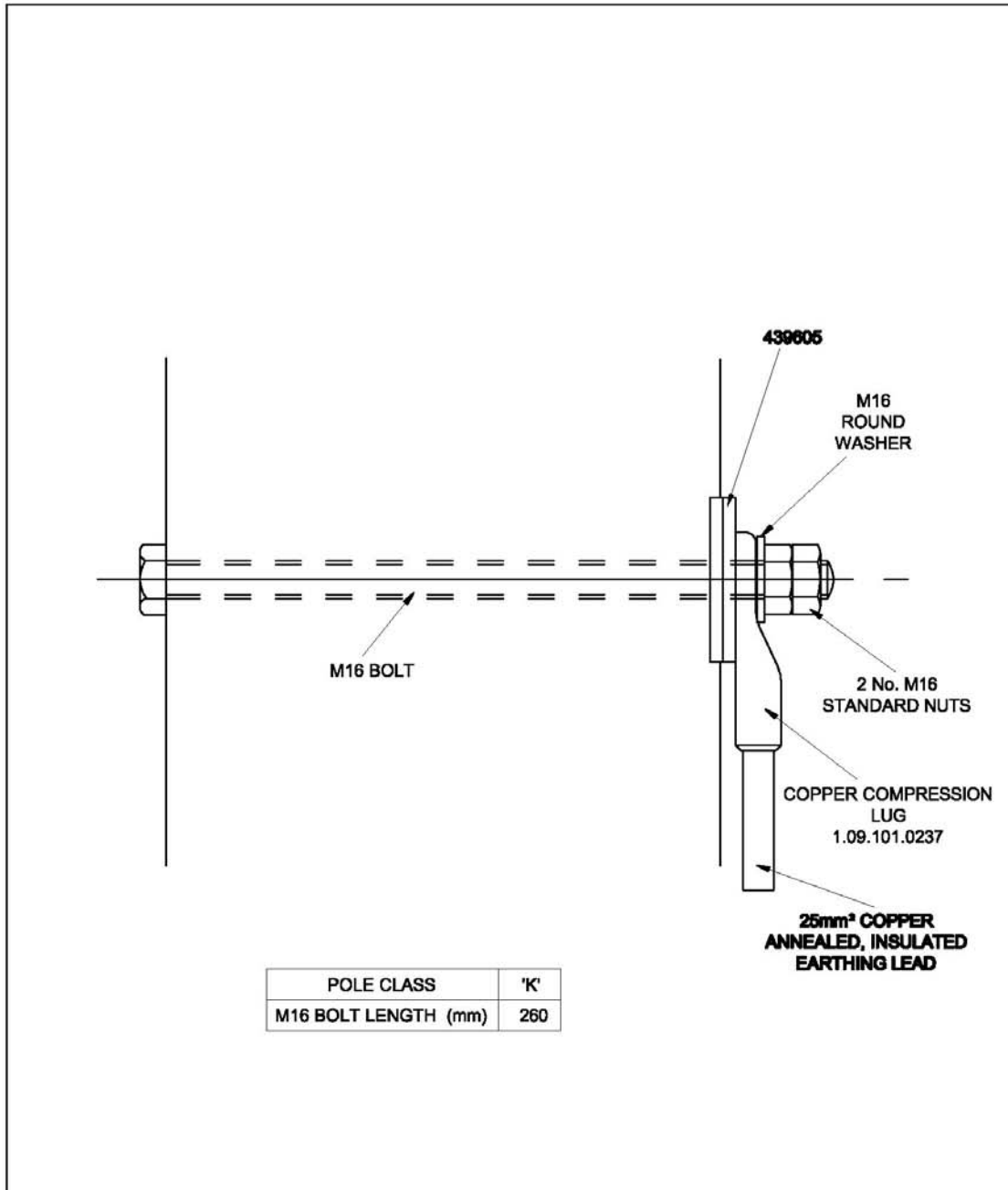


ALL DIMENSIONS SHOWN ARE IN MILLIMETRES UNLESS OTHERWISE STATED

WORK SPEC
No 04 /404

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		ARRANGEMENT OF EARTHING CONNECTION FOR STEEL POLES			
Sheet No. 1	Scale				
Prepared By Barbara.Simpson		Type OVERHEAD			
Revised 18/05/08	Grid Reference	Ref No.	C319036	Historic Drg.No.	1091193622
Date Issued 10/06/05	Checked By G.HAMMEL	Revision	C	Notes	

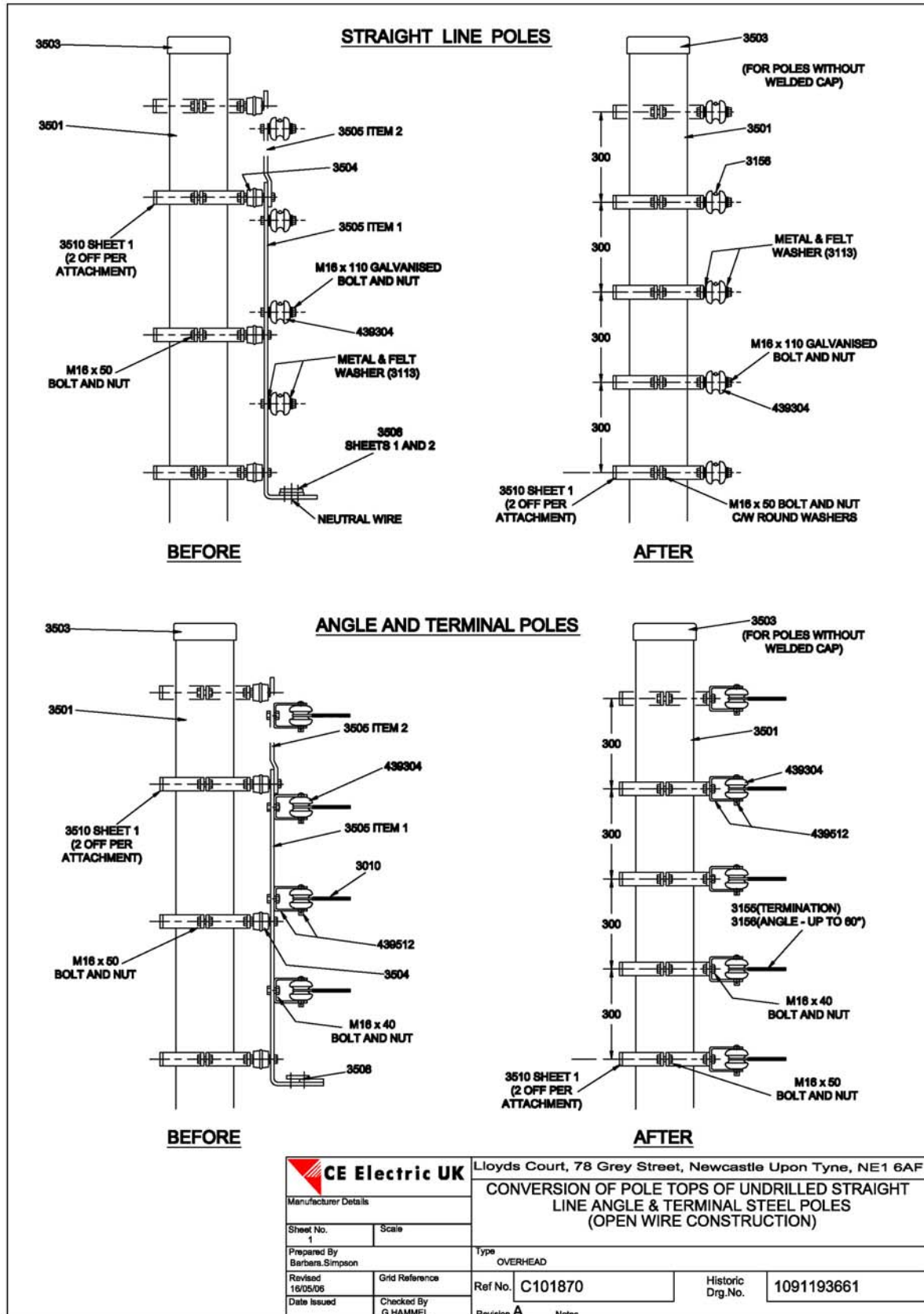
Document reference		NSP/004 /041/001			
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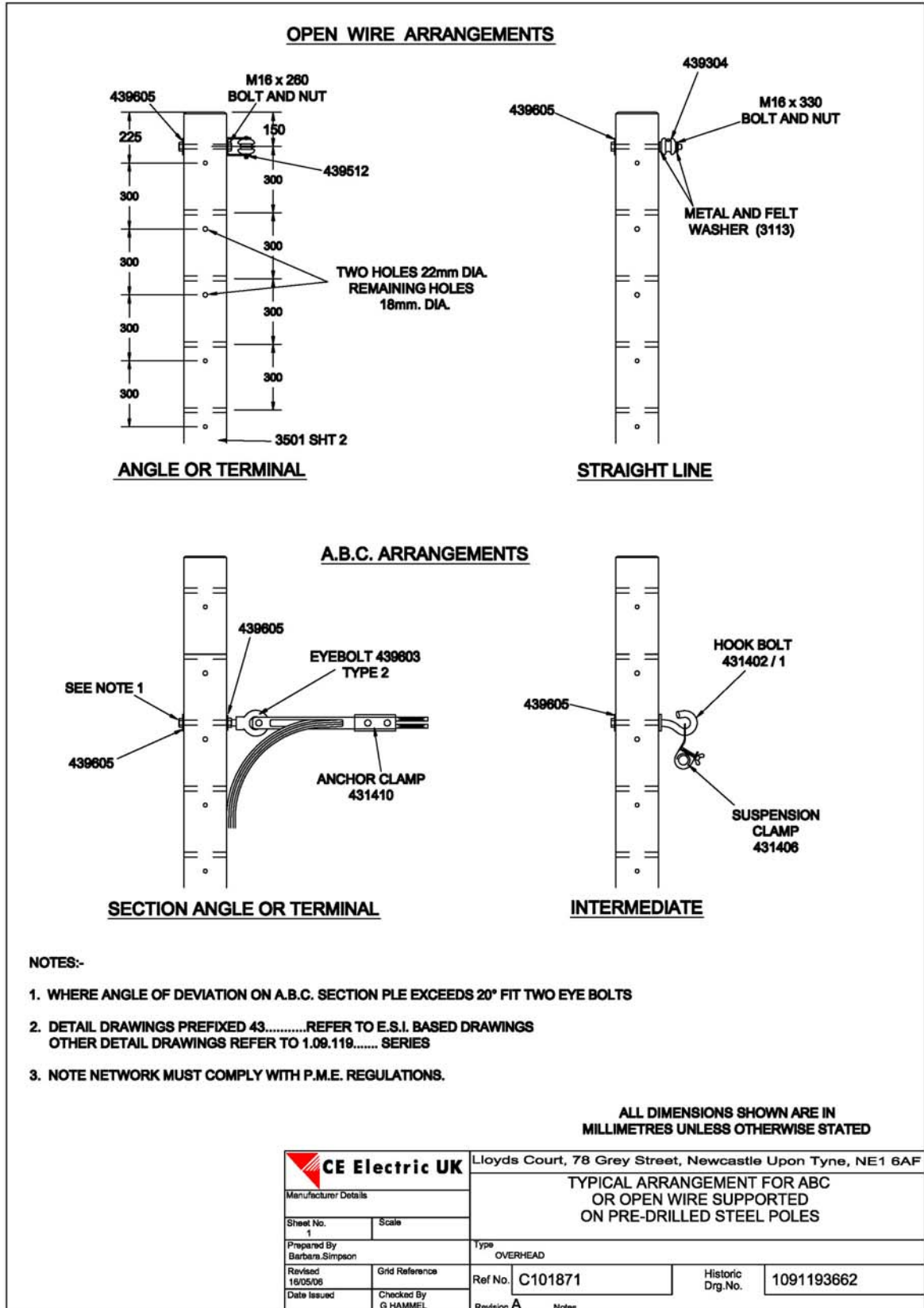
POLE CLASS	'K'
M16 BOLT LENGTH (mm)	260

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		LV TUBULAR STEEL POLES BONDING CONNECTION AT TOP OF POLES CLASSES K	
Sheet No. 1	Scale	Type OVERHEAD	
Prepared By Barbara Simpson	Grid Reference	Ref No. C101864	Historic Drg.No. 1091193525
Revised 18/05/08	Checked By G HAMMEL	Revision B	Notes
Date Issued 07/06/05			

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Appendix 2 – Openwire Network Conductor erection charts

ERECTOR AND DESIGN SAG AND TENSION CHART. 32 mm² (3/3.75 mm) H.D.B.C.

TEMP.
°C

TENSION.

SAG FOR VARIOUS SPANS.
(METRES)

kgf

NEWTONS

20

25

30

35

40

45

-5.6

378

3708

0.04

0.06

0.09

0.12

0.16

0.20

0

342

3354

0.04

0.07

0.10

0.13

0.17

0.22

5

311

3049

0.05

0.07

0.11

0.15

0.19

0.24

10

281

2755

0.05

0.08

0.12

0.16

0.21

0.27

15

253

2478

0.06

0.09

0.13

0.18

0.23

0.30

20

226

2220

0.07

0.10

0.15

0.20

0.26

0.33

25

203

1988

0.07

0.11

0.16

0.22

0.29

0.37

30

182

1782

0.08

0.13

0.18

0.25

0.33

0.41

50

124

1218

0.12

0.19

0.27

0.37

0.48

0.61

Prestress to
1/2 the Sag
at 20°C.

TABLE 1. - Normal tension.

M.W.T. 4450 N (454 kgf.)
F.O.S. 2.86.

TEMP.
°C

TENSION.

SAG FOR VARIOUS SPANS.
(METRES)

kgf

NEWTONS

20

25

30

35

40

45

-5.6

165

1619

0.09

0.14

0.20

0.28

0.36

0.46

0

148

1448

0.10

0.16

0.23

0.31

0.40

0.51

5

135

1321

0.11

0.17

0.25

0.34

0.44

0.56

10

124

1215

0.12

0.19

0.27

0.37

0.48

0.61

15

115

1125

0.13

0.20

0.29

0.40

0.52

0.65

20

107

1050

0.14

0.22

0.31

0.42

0.55

0.70

25

100

985

0.15

0.23

0.33

0.45

0.59

0.75

30

95

929

0.16

0.24

0.35

0.48

0.63

0.79

50

78

769

0.19

0.30

0.43

0.58

0.76

0.96

Prestress to
1/2 the Sag
at 0°C.

TABLE 2. - Reduced tension
for steel poles.

M.W.T. 2890 N (295 kgf.)
F.O.S. 4.40.

U.T.S. 12710 N (1296 kgf)
F.O.S. } As specified on
M.W.T. } individual tables.
LDG. WIND. 380 N/m²; ICE 9.5 mm.
NOM. DIA. 8.05 mm.
WT. PER m 0.297 kg.

C.O. EXP.
MOD. ELAS.
BASIC SPAN.

17 x 10⁻⁶ /°C.
124 x 10³ N/mm².
40 m.

NOTE :-

Tables 1 and 2 are the metric
equivalents of Tables 1 and 2
of D.H.L. 3/70.

NORTHERN ELECTRIC

DRG. No. 1.09.140.3404

SHEET No. 1A No. OF SHEETS TO DRG.

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Appendix 3 - Openwire Network Conductor erection charts

ERECTION AND DESIGN SAG AND TENSION CHART.

32mm² (3/3.75mm) H.O. P.V.C.
(TYPE 8) COPPER.

BASIC SPAN - 40 m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)					
	kgf	NEWTONS	20	25	30	35	40	45
- 5.6	357	3501	0.05	0.08	0.11	0.15	0.20	0.25
0	323	3168	0.05	0.08	0.12	0.17	0.22	0.27
5	294	2885	0.06	0.09	0.13	0.18	0.24	0.30
10	267	2618	0.07	0.10	0.15	0.20	0.26	0.33
15	242	2372	0.07	0.11	0.16	0.22	0.29	0.37
20	219	2147	0.08	0.12	0.18	0.24	0.30	0.40
25	199	1948	0.09	0.14	0.20	0.27	0.35	0.45
30	181	1774	0.10	0.15	0.22	0.30	0.39	0.49
50	132	1290	0.13	0.21	0.30	0.41	0.53	0.67

Prestress to
1/2 the Sag
at 20°C.

TABLE 1. - Normal tension.

M.W.T. 4450 N (454 kgf.)
F.O.S. 2.86.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)					
	kgf	NEWTONS	20	25	30	35	40	45
- 5.6	151	1483	0.12	0.18	0.26	0.35	0.46	0.59
0	139	1362	0.13	0.20	0.28	0.39	0.50	0.64
5	130	1271	0.14	0.21	0.30	0.41	0.54	0.68
10	122	1193	0.14	0.22	0.32	0.44	0.58	0.73
15	115	1125	0.15	0.24	0.34	0.47	0.61	0.77
20	109	1066	0.16	0.25	0.36	0.49	0.64	0.82
25	103	1014	0.17	0.26	0.38	0.52	0.68	0.86
30	99	968	0.18	0.28	0.40	0.54	0.71	0.90
50	84	828	0.21	0.32	0.47	0.63	0.83	1.05

Prestress to
1/2 the Sag
at 20°C.

TABLE 2. - Reduced tension
for steel poles.

M.W.T. 2890 N (295 kgf.)
F.O.S. 4.40.

BASIC DESIGN CRITERIA.

U.T.S. 12710 N (1296 kgf.)
F.O.S. } As specified on
M.W.T. } individual tables.
LDG. WIND. 380 N / m², ICE 9.5 mm.
NOM. DIA. 10.5 mm.
WT. PER m. 0.35 kg.

C.O. EXP.
MOD. ELAS.
BASIC SPAN.

17 x 10⁻⁶ /°C.
124 x 10³ N/mm².
40 m.

NOTE

NORTHERN ELECTRIC

DRG. No. 1, 09, 140, 3404

SHEET No. 2A No. OF SHEETS TO DRG.

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Appendix 4- Openwire Network Conductor erection charts

ERECTION AND DESIGN SAG AND TENSION CHART. 70mm² (7/3.55mm) HD.B.C.

BASIC SPAN - 40m.

TEMP °C.	TENSION.		SAG FOR VARIOUS SPANS. (METRES)					
	kgf	NEWTONS	20	25	30	35	40	45
- 5.6	499	4898	0.06	0.10	0.14	0.19	0.25	0.31
0	441	4328	0.07	0.11	0.16	0.22	0.28	0.36
5	395	3876	0.08	0.12	0.18	0.24	0.31	0.40
10	355	3482	0.09	0.14	0.20	0.27	0.35	0.44
15	321	3146	0.10	0.15	0.22	0.30	0.39	0.49
20	292	2862	0.11	0.17	0.24	0.33	0.43	0.54
25	268	2623	0.12	0.18	0.26	0.36	0.46	0.59
30	247	2423	0.13	0.20	0.28	0.38	0.50	0.64
50	192	1881	0.16	0.25	0.36	0.50	0.65	0.82

Prestress to 1/2 the Sag at 20°C.

TABLE 1. - Normal Tension.

M.W.T. 6000 N (612 kgf.)
F.O.S. 4.48.

TEMP °C	TENSION.		SAG FOR VARIOUS SPAN. (METRES)					
	kgf	NEWTONS	20	25	30	35	40	45
- 5.6	173	1697	0.18	0.28	0.40	0.55	0.72	0.91
0	165	1618	0.19	0.29	0.42	0.58	0.75	0.95
5	159	1555	0.20	0.31	0.44	0.60	0.78	0.99
10	153	1498	0.20	0.32	0.46	0.62	0.81	1.03
15	148	1447	0.21	0.33	0.47	0.64	0.84	1.07
20	143	1401	0.22	0.34	0.49	0.67	0.87	1.10
25	138	1358	0.22	0.35	0.50	0.69	0.90	1.14
30	135	1319	0.23	0.36	0.52	0.71	0.92	1.17
50	121	1190	0.26	0.40	0.58	0.78	1.02	1.30

Prestress to 1/2 the Sag at 15°C.

TABLE 2. - Reduced tension for steel poles.

M.W.T. 2890 N (295 kgf.)
F.O.S. 9.30.

U.T.S. 26880N (2741 kgf)
F.O.S. } As specified on individual tables.
M.W.T. }

LDG. WIND 380 N/m²; ICE 9.5mm.
NOM. DIA. 10.65mm.
WT. PER m. 0.621kg.

C.O. EXP.
MOD. ELAS.
BASIC SPAN.

17 x 10⁻⁶ /°C.
124 x 10³ N/mm².
40m.

NOTE. Tables 1 and 2 are the metric equivalents of Tables 1 and 2 of O.H.L. 3/70.

BASIC DESIGN CRITERIA.

NORTHERN ELECTRIC

DRG. No. 1, 09, 140, 3405
SHEET No. 1A No. OF SHEETS TO DRG.

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Appendix 5 Openwire Network Conductor erection charts

ERECTION AND DESIGN SAG AND TENSION CHART.
(TYPE B) COPPER.
70mm² (7/3.55mm) H.D.P.V.C.

BASIC SPAN - 40 m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)						
	kgf	NEWTONS	20	25	30	35	40	45	
- 5.6	474	4651	0.07	0.11	0.16	0.22	0.29	0.37	
0	423	4146	0.08	0.13	0.17	0.25	0.33	0.41	
5	383	3752	0.09	0.14	0.20	0.28	0.36	0.46	
10	348	3412	0.10	0.16	0.22	0.30	0.40	0.50	
15	318	3122	0.11	0.17	0.24	0.33	0.43	0.55	
20	293	2874	0.12	0.18	0.27	0.36	0.47	0.60	
25	272	2664	0.13	0.20	0.29	0.39	0.51	0.64	
30	254	2486	0.14	0.21	0.31	0.42	0.55	0.69	
50	202	1984	0.17	0.27	0.38	0.52	0.68	0.86	

Prestress to
1/2 the Sag
at 20°C.

TABLE 1. - Normal tension.

M.W.T. 6000 N (612 kgf.)
F.O.S. 4.48.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)						
	kgf	NEWTONS	20	25	30	35	40	45	
- 5.6	167	1638	0.21	0.32	0.47	0.63	0.83	1.05	
0	161	1578	0.21	0.34	0.48	0.66	0.86	1.09	
5	156	1529	0.22	0.35	0.50	0.68	0.89	1.12	
10	151	1484	0.23	0.36	0.51	0.70	0.91	1.16	
15	147	1443	0.23	0.37	0.53	0.72	0.94	1.19	
20	143	1405	0.24	0.38	0.54	0.74	0.96	1.22	
25	140	1370	0.25	0.39	0.56	0.76	0.99	1.25	
30	136	1337	0.25	0.40	0.57	0.78	1.01	1.28	
50	125	1225	0.28	0.43	0.62	0.85	1.11	1.40	

Prestress to
1/2 the Sag
at 10°C.

TABLE 2. - Reduced tension
for steel poles.

M.W.T. 2890 N (295 kgf.)
F.O.S. 9.30.

BASIC DESIGN CRITERIA.

U.V.S. 26880 N (2741 kgf.)
F.O.S. } As specified on
M.W.T. } individual tables.
LDG. WIND. 380 N/m², ICE 9.5 mm.
NOM. DIA. 13.5 mm.
WT. PER m. 0.691 kg.

C.O. EXP.
MOD. ELAS.
BASIC SPAN.

17 x 10⁻⁶ /°C.
124 x 10³ N/mm².
40 m.

NORTHERN ELECTRIC.

DRG. No. 1, 09, 140, 3405

SHEET No. 2A No. OF SHEETS TO DRG.

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Appendix 6 Openwire Network Conductor erection charts

ERECTION AND DESIGN SAG AND TENSION CHART.

50mm² (7/3.10mm)
PLAIN ALUMINIUM.

BASIC SPAN. - 40 m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	351	3439	0.02	0.03	0.05	0.06	0.08	0.11	0.13	0.16	0.19	
0	311	3048	0.02	0.04	0.05	0.07	0.09	0.12	0.15	0.18	0.21	
5	276	2702	0.03	0.04	0.06	0.08	0.11	0.13	0.17	0.20	0.24	
10	241	2361	0.03	0.05	0.07	0.09	0.12	0.15	0.19	0.23	0.27	
15	207	2030	0.04	0.06	0.08	0.11	0.14	0.18	0.22	0.27	0.32	
20	175	1713	0.04	0.07	0.09	0.13	0.17	0.21	0.26	0.32	0.38	
25	145	1420	0.05	0.08	0.11	0.15	0.20	0.26	0.31	0.38	0.45	
30	119	1164	0.06	0.10	0.14	0.19	0.25	0.31	0.38	0.46	0.55	
50	61	599	0.12	0.19	0.27	0.37	0.48	0.60	0.75	0.90	1.07	

Prestress to
1/2 the Sag
at 15°C.

TABLE 1. - Normal tensions.

M.W.T. 4150 N (423 kgf.)

F.O.S. 2.00.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	266	2608	0.03	0.04	0.06	0.08	0.11	0.14	0.17	0.21	0.25	
0	227	2229	0.03	0.05	0.07	0.10	0.13	0.16	0.20	0.24	0.29	
5	194	1903	0.04	0.06	0.08	0.12	0.15	0.19	0.23	0.28	0.34	
10	163	1594	0.04	0.07	0.10	0.14	0.18	0.23	0.28	0.34	0.40	
15	134	1314	0.05	0.09	0.12	0.17	0.22	0.28	0.34	0.41	0.49	
20	110	1077	0.07	0.10	0.15	0.20	0.27	0.34	0.42	0.50	0.60	
25	91	889	0.08	0.13	0.18	0.25	0.32	0.41	0.50	0.61	0.72	
30	76	749	0.10	0.15	0.21	0.29	0.38	0.48	0.60	0.72	0.86	
50	48	471	0.15	0.24	0.34	0.46	0.61	0.77	0.95	1.15	1.37	

Prestress to
1/2 the Sag
at 10°C.

TABLE 2. - Reduced tension
for steel poles.

M.W.T. 3560 N (363 kgf.)

F.O.S. 2.33.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)					
	kgf	NEWTONS	20	25	30	35	40	45
-5.6	87	858	0.08	0.13	0.19	0.26	0.33	0.42
0	73	713	0.10	0.16	0.23	0.31	0.40	0.51
5	63	621	0.12	0.18	0.26	0.35	0.46	0.58
10	56	553	0.13	0.20	0.29	0.40	0.52	0.66
15	51	500	0.14	0.22	0.32	0.44	0.57	0.72
20	47	459	0.16	0.24	0.35	0.48	0.62	0.79
25	43	426	0.17	0.26	0.38	0.51	0.67	0.85
30	41	399	0.18	0.28	0.40	0.55	0.72	0.91
50	33	325	0.22	0.34	0.50	0.67	0.88	1.12

M.W.T. 2440 N (249 kgf.)
F.O.S. 3.39.

TABLE 3. - Reduced tensions for
steel poles where Table
2 tensions would exceed
S.W.L. of pole.

BASIC DESIGN CRITERIA.

U.T.S. 8280 N (844 kgf)

F.O.S. } As specified on

M.W.T. } individual tables.

LDG. WIND. 380 N/m², I.C.E 9.5 mm.

NOM. DIA. 9.3 mm.

W.T. PER M. 0.146 kg.

CO. EXP. 23 x 10⁻⁶ /°C.

MOD. ELAS. 59 x 10³ N/mm².

BASIC SPAN. 40 m.

NOTE - Tables 1, 2 and 3 are the
metric equivalents of Tables 3,
4 and 5 O.H.L. 3/70.

NORTHERN ELECTRIC

DRG. No. 1 09 140 3406

SHEET No. 1A No. OF SHEETS TO DRG.

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Appendix 7 Openwire Network Conductor erection charts

ERECTION AND DESIGN SAG AND TENSION CHART. (TYPE 8) PLAIN ALUMINIUM. 50mm² (7/3.10 mm) P.V.C.

BASIC SPAN. - 40 m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	330	3236	0.03	0.05	0.07	0.09	0.12	0.15	0.19	0.23	0.27	
0	291	2856	0.03	0.05	0.08	0.11	0.14	0.17	0.22	0.26	0.31	
5	258	2525	0.04	0.06	0.09	0.12	0.16	0.20	0.24	0.29	0.35	
10	225	2206	0.04	0.07	0.10	0.14	0.18	0.23	0.28	0.34	0.40	
15	194	1904	0.05	0.08	0.12	0.16	0.21	0.26	0.32	0.39	0.47	
20	166	1627	0.06	0.09	0.14	0.19	0.24	0.31	0.38	0.46	0.54	
25	141	1385	0.07	0.11	0.16	0.22	0.28	0.36	0.44	0.54	0.64	
30	121	1183	0.08	0.13	0.19	0.26	0.33	0.42	0.52	0.63	0.75	
50	74	721	0.14	0.21	0.31	0.42	0.55	0.69	0.85	1.03	1.23	

Prestress to
1/2 the Sag
at 15°C.

TABLE 1. - Normal tension.

M.W.T. 4150 N (423 kgf.)
F.O.S. 2.00.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	240	2358	0.04	0.07	0.09	0.13	0.17	0.21	0.26	0.32	0.38	
0	205	2011	0.05	0.08	0.11	0.15	0.20	0.25	0.31	0.37	0.44	
5	176	1724	0.06	0.09	0.13	0.17	0.23	0.29	0.36	0.43	0.51	
10	150	1468	0.07	0.10	0.15	0.21	0.27	0.34	0.42	0.51	0.60	
15	128	1251	0.08	0.12	0.18	0.24	0.31	0.40	0.49	0.60	0.71	
20	110	1075	0.09	0.14	0.21	0.28	0.37	0.46	0.57	0.69	0.82	
25	96	937	0.11	0.16	0.24	0.32	0.42	0.53	0.66	0.80	0.95	
30	85	830	0.12	0.19	0.27	0.36	0.47	0.60	0.74	0.90	1.07	
50	60	584	0.17	0.26	0.38	0.52	0.67	0.85	1.05	1.27	1.52	

Prestress to
1/2 the Sag
at 5°C.

TABLE 2. - Reduced tension for
steel poles.

M.W.T. 3560 N (363 kgf.)
F.O.S. 2.33.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)						
	kgf	NEWTONS	20	25	30	35	40	45	
-5.6	82	808	0.12	0.19	0.27	0.37	0.49	0.62	
0	74	721	0.14	0.21	0.31	0.42	0.55	0.69	
5	67	660	0.15	0.23	0.34	0.46	0.60	0.76	
10	62	611	0.16	0.25	0.36	0.49	0.64	0.82	
15	58	571	0.17	0.27	0.39	0.53	0.69	0.87	
20	55	536	0.18	0.29	0.41	0.56	0.73	0.93	
25	52	507	0.19	0.30	0.44	0.59	0.78	0.98	
30	49	482	0.20	0.32	0.46	0.63	0.82	1.03	
50	42	409	0.24	0.38	0.54	0.74	0.96	1.22	

TABLE 3. - Reduced tensions for steel
poles where Table 2
tensions would exceed
S.W.L. of pole.

M.W.T. 2440 N (249 kgf.)
F.O.S. 3.39.

BASIC DESIGN CRITERIA.

U.T.S. 8280 N (844 kgf.)

F.O.S. } As specified on

M.W.T. } individual tables.

LDG. WIND. 380 N / m²; ICE 9.5 mm.

NOM. DIA. 11.7 mm.

W.T. PER m. 0.201 kg.

CO. EXP. 23 x 10⁻⁶ / °C.

MOD. ELAS. 59 x 10³ N / mm².

BASIC SPAN. 40 m.

NORTHERN ELECTRIC

DRG. No. 1 109,140 3406

SHEET No. 2A No. 0 SHEETS TO DRG.

NORTHERN ELECTRIC

DRG. No. 1, 09, 140, 3406
SHEET No. 2A No. OF SHEETS TO DRG.

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Appendix 8 Openwire Network Conductor erection charts

100mm² (7/4.39mm)
PLAIN ALUMINIUM.

ERECTION AND DESIGN SAG AND TENSION CHART.

BASIC SPAN. - 40m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	506	4962	0.03	0.04	0.06	0.09	0.12	0.15	0.18	0.22	0.26	
0	429	4210	0.03	0.05	0.08	0.10	0.14	0.17	0.21	0.26	0.31	
5	364	3567	0.04	0.06	0.09	0.12	0.16	0.20	0.25	0.30	0.36	
10	303	2967	0.05	0.08	0.11	0.15	0.19	0.24	0.30	0.36	0.43	
15	248	2435	0.06	0.09	0.13	0.18	0.23	0.30	0.37	0.44	0.53	
20	203	1995	0.07	0.11	0.16	0.22	0.29	0.36	0.45	0.54	0.64	
25	169	1658	0.09	0.13	0.19	0.26	0.34	0.44	0.54	0.65	0.77	
30	144	1410	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.77	0.91	
50	93	912	0.16	0.24	0.35	0.48	0.63	0.79	0.98	1.18	1.41	

Prestress to
1/2 the Sag
at 10°C.

TABLE 1. - Normal tension.

M.W.T. 6000 N (612 kgf.)
F.O.S. 2.67.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	216	2113	0.07	0.11	0.15	0.21	0.27	0.34	0.42	0.51	0.61	
0	174	1709	0.08	0.13	0.19	0.26	0.33	0.42	0.52	0.63	0.75	
5	148	1448	0.10	0.15	0.22	0.30	0.39	0.50	0.62	0.75	0.89	
10	128	1257	0.11	0.18	0.26	0.35	0.45	0.57	0.71	0.86	1.02	
15	114	1116	0.13	0.20	0.29	0.39	0.51	0.65	0.80	0.97	1.15	
20	103	1009	0.14	0.22	0.32	0.43	0.57	0.72	0.88	1.07	1.27	
25	94	924	0.15	0.24	0.35	0.47	0.62	0.78	0.97	1.17	1.39	
30	87	857	0.17	0.26	0.37	0.51	0.67	0.84	1.04	1.26	1.50	
50	69	681	0.21	0.33	0.47	0.64	0.84	1.06	1.31	1.59	1.85	

Prestress to
1/2 the Sag
at 0°C.

TABLE 2. - Reduced tension
for steel pole.

M.W.T. 4000 N (408 kgf.)
F.O.S. 4.00.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)						
	kgf	NEWTONS	20	25	30	35	40	45	
-5.6	92	902	0.16	0.25	0.36	0.48	0.63	0.80	
0	85	832	0.17	0.27	0.39	0.53	0.69	0.87	
5	80	780	0.18	0.29	0.41	0.56	0.73	0.93	
10	75	737	0.19	0.30	0.44	0.59	0.78	0.98	
15	71	700	0.20	0.32	0.46	0.62	0.82	1.03	
20	68	667	0.21	0.33	0.48	0.66	0.86	1.08	
25	65	639	0.22	0.35	0.50	0.68	0.89	1.13	
30	63	614	0.23	0.36	0.52	0.71	0.93	1.18	
50	55	536	0.27	0.42	0.60	0.82	1.07	1.35	

TABLE 3. - Reduced tensions for
steel poles where Table
2 tensions would exceed
SW.L. of pole.

M.W.T. 2670 N (273 kgf.)
F.O.S. 6.00.

BASIC DESIGN CRITERIA.

U.T.S. 16000 N (1631 kgf.)

F.O.S. } As specified on

M.W.T. } individual tables.

LDG. WIND 380 N / m², ICE 9.5mm.

NOM. DIA. 13.17mm.

W.T. PER m. 0.291kg.

CO. EXP. 23 x 10⁻⁶ /°C

MOD. ELAS. 59 x 10³ N / mm².

BASIC SPAN. 40m.

NOTE :- Tables 1, 2 and 3 are the
metric equivalents of Tables 3, 4 and
5 of O.H.L. 3/70.

NORTHERN ELECTRIC

DRG. No. 1 09,140 3407

SHEET No.1A No. OF SHEETS TO DRG.

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Appendix 9 Openwire Network Conductor erection charts

ERECTOR AND DESIGN SAG AND TENSION CHART. 100mm² (7/4.39 mm) P.V.C. (TYPE 8) PLAIN ALUMINIUM.

BASIC SPAN. - 40 m.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	478	4691	0.04	0.06	0.08	0.12	0.15	0.19	0.24	0.29	0.34	
0	406	3979	0.04	0.07	0.10	0.14	0.18	0.23	0.28	0.34	0.40	
5	345	3386	0.05	0.08	0.12	0.16	0.21	0.26	0.33	0.40	0.47	
10	291	2853	0.06	0.10	0.14	0.19	0.25	0.31	0.39	0.47	0.56	
15	244	2399	0.07	0.12	0.17	0.23	0.30	0.37	0.46	0.56	0.66	
20	207	2035	0.09	0.14	0.20	0.27	0.35	0.44	0.54	0.66	0.78	
25	179	1753	0.10	0.16	0.23	0.31	0.40	0.51	0.63	0.76	0.91	
30	157	1539	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87	1.04	
50	109	1065	0.17	0.26	0.37	0.51	0.67	0.84	1.04	1.26	1.50	

Prestress to 1/2 the Sag at 10°C.

TABLE 1. - Normal tension.

M.W.T. 6000 N (612 kgf.)

F.O.S. 2.67.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)									
	kgf	NEWTONS	20	25	30	35	40	45	50	55	60	
-5.6	197	1927	0.09	0.14	0.21	0.28	0.37	0.47	0.57	0.69	0.83	
0	168	1645	0.11	0.17	0.24	0.33	0.43	0.54	0.67	0.81	0.97	
5	149	1457	0.12	0.19	0.27	0.37	0.49	0.62	0.76	0.92	1.09	
10	134	1312	0.14	0.21	0.30	0.41	0.54	0.68	0.84	1.02	1.22	
15	122	1198	0.15	0.23	0.33	0.45	0.59	0.75	0.92	1.12	1.33	
20	113	1106	0.16	0.25	0.36	0.49	0.64	0.81	1.00	1.21	1.44	
25	105	1031	0.17	0.27	0.39	0.53	0.69	0.87	1.07	1.30	1.55	
30	99	968	0.18	0.29	0.41	0.56	0.73	0.93	1.14	1.38	1.65	
50	81	795	0.22	0.35	0.50	0.68	0.89	1.13	1.39	1.69	2.01	

Prestress to 1/2 the Sag at - 5.6°C.

TABLE 2. - Reduced tension for steel poles.

M.W.T. 4000 (408 kgf.)

F.O.S. 4.00.

TEMP. °C	TENSION.		SAG FOR VARIOUS SPANS. (METRES)						
	kgf	NEWTONS	20	25	30	35	40	45	
-5.6	95	928	0.19	0.30	0.43	0.58	0.76	0.97	
0	89	875	0.20	0.32	0.46	0.62	0.81	1.02	
5	85	834	0.21	0.33	0.48	0.65	0.85	1.07	
10	81	799	0.22	0.35	0.50	0.68	0.89	1.12	
15	78	767	0.23	0.36	0.52	0.71	0.92	1.17	
20	75	739	0.24	0.37	0.54	0.73	0.96	1.21	
25	73	714	0.25	0.39	0.56	0.76	0.99	1.26	
30	70	691	0.26	0.40	0.58	0.79	1.03	1.30	
50	63	616	0.29	0.45	0.65	0.88	1.15	1.46	

M.W.T. 2670 N (273 kgf.)

F.O.S. 6.00.

TABLE 3. - Reduced tensions for steel poles where Table 2 tensions would exceed SWL of pole.

BASIC DESIGN CRITERIA.

U.T.S. 16000 N (1631 kgf)

F.O.S. } As specified on

M.W.T. } individual tables.

LDG. WIND. 380 N/m², ICE 9.5 mm.

NOM. DIA. 16 mm.

W.T. PER m. 0.361 kg.

CO. EXP. 23 x 10⁻⁶/°C.

MOD. ELAS. 59 x 10³ N/mm².

BASIC SPAN. 40 m.

NORTHERN ELECTRIC

DRG. No. 1 09 140 3407

SHEET No. 2A NO. OF SHEETS TO DRS

NORTHERN ELECTRIC
 DRG. No. 1 09 140 3407
 SHEET No. 2A No. OF SHEETS TO DRG.

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Appendix 10 - Minimum Stay Spreads For Angle Poles

4 x 32 mm² HDBC

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.0	1@4.9	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.4
9.5	1@4.3	1@4.3	1@4.7	1@5.4	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.7
10.0	1@4.6	1@4.6	1@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@5.0
10.5	1@4.9	1@5.3	1@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@5.3
11.0	1@5.2	1@6.3	1@5.2	1@6.3	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.6
12.0	1@5.8	1@6.7	1@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@6.0	2@6.5	2@7.0

4 x 70 mm² HDBC

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.0	1@4.7	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.4
9.5	1@4.3	1@4.3	1@4.9	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.7
10.0	1@4.6	1@4.8	1@5.8	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@5.0
10.5	1@4.9	1@5.7	1@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@5.3
11.0	1@5.4	1@5.2	1@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.6
12.0	1@7.2	1@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@6.2	2@6.7	2@7.2

4 x 100 mm² HDBC

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.2	1@5.1	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.4
9.5	1@4.3	1@4.3	1@5.0	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.7
10.0	1@4.6	1@5.0	1@4.8	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@5.0
10.5	1@4.9	1@4.9	1@5.1	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@5.3
11.0	1@5.8	1@5.2	1@5.4	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.6
12.0	1@5.8	1@5.8	1@6.0	2@5.8	2@5.8	2@5.8	2@5.8	2@6.2	2@7.0	2@7.2

	Medium Grade
	Stout Grade

Notes

- The chart created to provide the best combination of pole grade / number of stays and minimum spread
- Alternative solutions are available by manually checking the strut loads and stay wire capabilities against ENA TS 43-30. Table 5 and the appropriate stay charts for the conductor size and number of wire.
- Based on 7/4.00mm grade 700 stay wire (Group 2 as defined in ENA TS 43-30 Table 6)
- Attachment point of stays at 600mm from pole top.
- Pole grade selected to keep stay spread below 35°
- Terminal loads shall be taken as equal to 60°

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Appendix 11 - Minimum Stay Spreads For Angle Poles

4 x 32 mm² PVC Covered Copper

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.0	1@4.0	1@4.2	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0
9.5	1@4.3	1@4.3	1@4.3	1@4.3	1@4.5	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3
10.0	1@4.6	1@4.6	1@4.6	1@4.8	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6
10.5	1@4.9	1@4.9	1@4.9	1@5.5	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9
11.0	1@5.2	1@5.2	1@5.6	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2
12.0	1@5.8	1@6.2	1@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@5.8

4 x 70 mm² PVC Covered Copper

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.2	1@5.1	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.4
9.5	1@4.3	1@4.3	1@5.0	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.2	2@4.7
10.0	1@4.6	1@5.0	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@5.0
10.5	1@4.9	1@5.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@5.3
11.0	1@5.2	1@5.2	1@5.2	1@5.2	1@5.2	1@5.2	1@5.2	1@5.2	1@5.2	2@5.6
12.0	1@7.5.8	1@5.8	1@6.0	2@5.8	2@5.8	2@5.8	2@5.8	2@6.2	2@6.7	2@7.2

4 x 100 mm² PVC Covered Copper

Dev	<15°	20°	25°	30°	35°	40°	45°	50°	55°	60°
Pole Height (m)	Minimum stay spread (m)									
9.0	1@4.0	1@4.0	1@4.4	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.0	2@4.4
9.5	1@4.3	1@4.3	1@5.2	1@4.5	2@4.3	2@4.3	2@4.3	2@4.3	2@4.3	2@4.7
10.0	1@4.6	1@5.2	1@4.8	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@4.6	2@5.0
10.5	1@4.9	1@4.9	1@5.1	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@4.9	2@5.3
11.0	1@6.1	1@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.2	2@5.6
12.0	1@5.8	1@5.8	2@5.8	2@5.8	2@5.8	2@5.8	2@6.0	2@6.5	2@7.0	2@7.5

	Medium Grade
	Stout Grade

Notes

- The chart created to provide the best combination of pole grade / number of stays and minimum spread
 - Alternative solutions are available by manually checking the strut loads and stay wire capabilities against ENA TS 43-30. Table 5 and the appropriate stay charts for the conductor size and number of wire.
 - Based on 7/4.00mm grade 700 stay wire (Group 2 as defined in ENA TS 43-30 Table 6)
 - Attachment point of stays at 600mm from pole top.
 - Pole grade selected to keep stay spread below 35°
- Terminal loads shall be taken as equal to 60°

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Appendix 12 - Historical Pole Type Classification Table

Length of Pole			Light			Medium			Medium Stout			Stout		
Approx. Conv. In (M)	Feet	Range of Pole Sink Depths (M)	Dia. at top (mm)		Min Dia 5ft from Butt (mm)	Dia. At top (mm)		Min Dia 5ft from Butt (mm)	Dia. at top (mm)		Min Dia 5ft from Butt (mm)	Dia. at top (mm)		Min Dia 5ft from Butt (mm)
			Min	Max		Min	Max		Min	Max		Min	Max	
8.5	28	1.5 - 1.8	127	152	178	146	178	216	158	203	241	190	241	266
9.0	30	1.5 - 1.8	127	152	184	152	184	222	158	209	247	190	241	273
10.0	32	1.5 - 1.8	127	158	184	152	184	229	165	209	254	190	247	279
10.5	34	1.5 - 1.8	127	158	184	152	184	235	165	209	260	190	247	285
11.0	36	1.5 - 1.8	127	165	184	152	184	241	165	216	266	190	247	292
11.5	38	1.8	127	165	184	152	184	248	171	228	273	190	247	298
12.0	40	1.8	127	165	203	152	184	248	171	228	273	190	247	305
13.0	42	1.8	-	-	-	159	197	260	178	228	285	190	247	317
14.0	45	2.1	133	171	222	165	203	273	178	228	298	197	254	330

Notes :

1. This table is only to be used where poles are not already classified with pole scarfings.

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Appendix 13 - Supplementary Guidance for ESQCR Refurbishment Schemes

The building clearance ESQCR programme is concerned solely with those assets which are affected by the rule change in 2002. That is, those LV mains that were OK under the old rules, being about 2m from a building, became unacceptable under the new rules, which retrospectively required a 3m clearance (imposed by a reference in the ESQCR guidelines to ENA-TS 43-8).

Therefore, only replacing LV mains which were sufficient under the old (2m) rule but become insufficient under the new (3m) rule can be charged to the building clearance ESQCR work programme. In practice, this will likely cover all LV mains with insufficient clearance.

In the process of this work, we will find a lot of other assets which are insufficient. This will include some 'ESQCR' issues, such as services with insufficient ground clearance, as well as routine deterioration such as VIR lead-ins. All of this work needs to be done and should form part of the overall refurbishment scheme. However, it is not driven by the 2002 rule change, and so should be allocated to general replacement work programs and not to the building clearance ESQCR work program.

When bare low voltage overhead lines are subjected to an ESQCR refurbishment the scope of the work shall be in accordance with this document with the following exemptions.

Main Line Bare Conductor Replacement

ESQCR requires that un-insulated LV network conductors within 3m of buildings or climbable trees are replaced with ABC or provided with some other form of permanent shrouding. Therefore the scope of this work is only to replace specific spans that infringe the 3m clearance requirement. In most cases this will result in a section to section conversion to ABC. (see clause 3.6.1 for details on the selected use of permanent LV shrouding as an alternative).

Pole Replacement

The condition of the poles only where the network conductors are to be replaced with ABC shall be considered for replacement. Section 3.5.3 of NSP/004/041/001 provides guidance on the wood pole condition assessment process with steel pole inspection criteria detailed in section 3.5.4. When replacing steel poles the possibility of replacing them with a wood pole shall be considered, for example straight line or light angle intermediate poles with only service tee offs.

Existing and replacement wood poles shall be a minimum medium grade with un-stayed light angles requiring stout grade poles. The grade of existing poles can be confirmed from the dimensional details below. shown in appendix 12.

Service Upgrade

Only services attached to the network poles that are to be converted to ABC should be considered for upgrade.

All bare neutral and Hessian covered aerals shall be replaced as they are not effectively insulated and cannot be terminated onto the ABC main. Bare or PVC covered Aluminium aerals shall also be replaced. Two, three or four wire PVC covered copper aerals do not require replacement. Some basic scenarios are detailed in the table below.

Existing Service – Pole Connection	Existing Service – Property Connections	Requirement
CNE - single and three phase	Any Scenario	No work required
ABC - two or 4 core	Any Scenario	No work required
PVC Copper Aerial - 2, 3 or 4 wire	Any Scenario	No work required
Hessian or Aerial with Bare Phase	Singe service	Replace aerial with concentric cable

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– 2 Wire		
Hessian or Aerial with Bare Phase – 2 Wire	Two or more services	Replace aerial with 2 or 4 core 35mm ABC
Hessian or Aerial with Bare Phase – 3 or 4 Wire	Any Scenario	Replace with 4 core 35mm ABC

It is permissible to retain branched services where the costs of replacing them by direct services would be significant. For example, if a two-wire service is to be replaced by a single concentric, it may be economic to install two concentric cables to split supplies to the premises involved.

Ground Clearance Issues

If ground clearance is infringed by services on networks where ABC is being erected thought shall be given to engineer a solution. This could include the relocating of service attachments or substituting open aerials for concentric or ABC services. The opportunity to re-site services currently on chimney brackets (that are not accessible via mobile access platforms) shall also be considered. Solutions may include the placement of a service pole or the redirecting of the service flight.

For poles in the proximity of walls and similar objects there shall be a minimum of 3m of foot-hold free surface to prevent unauthorised access. When this cannot be achieved an approved anti climbing device shall be fitted.

Other Network or Service Defects

When an ESQCR upgrade is in progress and solutions for particular design issues cannot be easily or cost effectively be overcome details shall be recorded and passed back to the Design Section.

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Appendix 14 - Wood Pole Condition Report Questions and Answers together with the associated “CR” rating

Notes:-

Table cells that are heavily greyed out contain condition ratings that are likely to require a pole to be replaced. All other lightly greyed out cells are condition ratings that must be resolved as part of any refurbishment works but are unlikely to cause the overall line condition rating to move from “Extensive Maintenance” or “Line Strengthening” into a “Line Rebuild”

Table Question No	Question	As found condition	“CR No
Q21	Pole Condition	Pole OK	1
Q21	Pole Condition	Minor Damage	2
Q21	Pole Condition	Decayed - OK > 1 Year	3
Q21	Pole Condition	Severely Decayed/Damaged	4
Q22	Boron Treatment	Not Boron Treated	1
Q22	Boron Treatment	Boron Treatment Recharged	2
Q22	Boron Treatment	Re-visit Req. Caps Accessible	3
Q22	Boron Treatment	Re-visit Req. Caps NOT Accessible	4
Q23	Pole Leaning	OK	1
Q23	Pole Leaning	Leaning < 1 Head	2
Q23	Pole Leaning	Leaning 1 -2 Heads	3
Q23	Pole Leaning	Leaning > 3 Heads	4
Q24	Scarf Mark Condition	OK @ Correct Height	1
Q24	Scarf Mark Condition	Mark Not Visible	3
Q24	Scarf Mark Condition	Mark too High	4
Q25	Steel Pole Thickness	Thickness > 95% of Original	1
Q25	Steel Pole Thickness	80 % - 90% of Original	2
Q25	Steel Pole Thickness	70% - 80% of Original	3
Q25	Steel Pole Thickness	< 70% of Original	4
Q28	ACD	No ACD Fitted to this Pole	1
Q28	ACD	ACD is OK	2
Q28	ACD	ACD/Barbed Wire Rusty but OK	3
Q28	ACD	ACD is Ineffective	4
Q29	Phase Colour Plates	Phase Colour Plates Not Changed	1
Q29	Phase Colour Plates	Phase Colour Plate Changed	2
Q29	Phase Colour Plates	No Phase Plates Fitted	3
Q30	DOD Condition	DOD OK to Specification	1
Q30	DOD Condition	DOD Faded but Legible	2
Q30	DOD Condition	Non Standard DOD Fitted	3
Q30	DOD Condition	DOD Unreadable or Missing	4
Q31	DOD Replacement	DOD Replaced && Old Removed	1
Q31	DOD Replacement	DOD Replace && Old Not Removed	2

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Q31	DOD Replacement	DOD Restricted && Old Removed	3
Q31	DOD Replacement	DOD Restricted && Old Not Removed	4
Q31	DOD Replacement	DOD Not Replaced	5
Q32	Property Plate Condition	Label OK	1
Q32	Property Plate Condition	Label Faded but Legible	2
Q32	Property Plate Condition	Label Damaged/Unreadable	3
Q32	Property Plate Condition	Label Missing	4
Q32	Property Plate Condition	Property Plate Not Required	5
Q33	Property Plate Replacement	Property Label Replaced	1
Q33	Property Plate Replacement	Property Label Not Replaced	2
Q33	Property Plate Replacement	Reason for Not Replacing Label	10
Q33	Property Plate Replacement	Not Required	3
Q34	Double Circuits	ID Plates OK	1
Q34	Double Circuits	ID Plates Faded but Legible	2
Q34	Double Circuits	ID Plates Damaged/Unreadable	3
Q34	Double Circuits	ID Plates Missing	4
Q34	Double Circuits	Single Circuit	5
Q35	Equipment Name Plate	Name Plate OK	1
Q35	Equipment Name Plate	Name Plate Faded but Legible	2
Q35	Equipment Name Plate	Name Plate Damaged/Unreadable	3
Q35	Equipment Name Plate	Name Plate Missing	4
Q35	Equipment Name Plate	No Equipment On Pole	5
Q36	Pole Number	Pole Number Replaced	1
Q36	Pole Number	Pole Num Replaced with Map Number	2
Q36	Pole Number	Pole Number NOT Replaced	3
Q36	Pole Number	Reason for not Replacing Number	10
Q37	Number of Pole Cables	Select Number of Cables	1
Q37_1	Cable Condition for Cable	Cable OK	1
Q37_1	Cable Condition for Cable	Cable Damaged but no Immediate Danger	2
Q37_1	Cable Condition for Cable	Cable Badly Deteriorated	3
Q37_1	Cable Condition for Cable	Cable Damaged. Danger to Public	4
Q38	Qty of Cable Guards that should be fitted	Select Number of Cable Guards	1
Q38_1	Cable Guard Condition for Guard	Cable Guard OK	1
Q38_1	Cable Guard Condition for Guard	Damaged but no Immediate Danger	2
Q38_1	Cable Guard Condition for Guard	Damage/Missing. Danger to Public	4
Q39	Number of Cable Boxes	Select Number of Cable Boxes	1
Q39_1	Cable Box Conditions for Box	Cable Box OK	1
Q39_1	Cable Box Conditions for Box	Box Rusty/Corroded	2
Q39_1	Cable Box Conditions for Box	Leaking Compound	3
Q39_1	Cable Box Conditions for Box	Arching/Burning at Cable Box	4
Q40	Quantity of Pole Earths	Select Number of Pole Earths	1
Q40_1	Earth Condition of Earth	Earth Wire OK	1

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Q40_1	Earth Condition of Earth	Damaged above 3m from Ground	2
Q40_1	Earth Condition of Earth	Damaged below 3m from Ground	3
Q40_1	Earth Condition of Earth	Earth Wire Severed/Missing	4
Q41	HV Earth/Op Handle Separation	Separation is Minimum 90 deg	1
Q41	HV Earth/Op Handle Separation	Separation is < 90 deg	4
Q41	HV Earth/Op Handle Separation	No Handle Fitted	5
Q42	Neutral/HV Earth Separation	Separation is Minimum 90 deg	1
Q42	Neutral/HV Earth Separation	Separation is < 90 deg	4
Q42	Neutral/HV Earth Separation	No Applicable	5
Q43	Number of Stays	Select Number of Stays	1
Q43_1	Wire Condition for Stay	Stay Wire OK	1
Q43_1	Wire Condition for Stay	Minor Corrosion	2
Q43_1	Wire Condition for Stay	Severely Corroded or Slack	3
Q43_1	Wire Condition for Stay	Broken or Missing	4
Q43_2	Insulator Condition for Stay	Stay Insulator OK	1
Q43_2	Insulator Condition for Stay	Insulator Damaged	2
Q43_2	Insulator Condition for Stay	Insulator too High or too Low	3
Q43_2	Insulator Condition for Stay	No Stay Insulator	4
Q43_3	Rod Condition for Stay	Stay Rod OK	1
Q43_3	Rod Condition for Stay	Rod Bent but OK	2
Q43_3	Rod Condition for Stay	Rod Severely Corroded	3
Q43_3	Rod Condition for Stay	Rod Broken or all below Ground	4
Q44	Number of Cross Arms	Select Number of Xarms	1
Q44_1	Condition of Crossarm	Steelwork OK	1
Q44_1	Condition of Crossarm	Light Rusting/Corrosion/Damage	2
Q44_1	Condition of Crossarm	Heavy Rusting/Corrosion/Damage	3
Q44_1	Condition of Crossarm	De-lamination Present or Badly Damaged	4
Q44_2	Orientation of Crossarm	Crossarm OK	1
Q44_2	Orientation of Crossarm	Crossarm Leaning	2
Q44_2	Orientation of Crossarm	Crossarm Twisted. Not at 90 deg	3
Q45	Angle of Deviation	Select Angle of Deviation	1
Q46	Condition of Fittings	Fittings OK	1
Q46	Condition of Fittings	Wear/Corrosion <25% Area	2
Q46	Condition of Fittings	Wear/Corrosion 25% - 75% Area	3
Q46	Condition of Fittings	Wear/Corrosion > 75% Area	4
Q47	Insulator Type	Pin Insulator	1
Q47	Insulator Type	Dish Insulator	2
Q47	Insulator Type	Bracket Type Insulator	3
Q47	Insulator Type	Other Type of Insulator	4
Q48	Condition of Insulators	Insulator(s) OK	1
Q48	Condition of Insulators	One Insulator Cracked/Chipped	2
Q48	Condition of Insulators	>1 Ins in String Cracked/Chip	3
Q48	Condition of Insulators	Insulator(s) Missing	4

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Q49	Insulator Orientation	Insulator(s) OK	1
Q49	Insulator Orientation	Pin Insulator Loose/Leaning	2
Q49	Insulator Orientation	String Pulled/Not Vertical	3
Q50	Condition Of Binders	Conductor Binders/Ties OK	1
Q50	Condition Of Binders	Conductor Binders/Ties Damaged	2
Q50	Condition Of Binders	Conductor Not Secure	4
Q50	Condition Of Binders	No Binders Present	5
Q51	Proximity Of Trees To Pole/Span	No Trees or Trees > 3.6m from Line	1
Q51	Proximity Of Trees To Pole/Span	Trees Within 2m - 3.6m of Line	2
Q51	Proximity Of Trees To Pole/Span	Trees Within 1m - 2m of Line	3
Q51	Proximity Of Trees To Pole/Span	Trees Within < 1m of Line	4
Q52	Proximity of Climbable Trees By Pole or Span	No Climbable Trees	1
Q52	Proximity of Climbable Trees By Pole or Span	Climbable Limbs within 2m -3.6m	2
Q52	Proximity of Climbable Trees By Pole or Span	Climbable Limbs within 1m - 2m	3
Q52	Proximity of Climbable Trees By Pole or Span	Climbable Limbs <1m or Touching	4
Q53	Third Party Interference	No Third Party Interference	1
Q53	Third Party Interference	Third Party Attachments on Pole	2
Q53	Third Party Interference	Evidence of Vandalism	4
Q61	Transformer	Pole Transformer OK	1
Q61	Transformer	Tx Tank > 50% Surface Rust	2
Q61	Transformer	Tx Bushing Damaged (HV or LV)	3
Q61	Transformer	Oil leaking from Tx	4
Q62	Transformer(2)	Tx Supporting Structure OK	1
Q62	Transformer(2)	Tx Support Rusty but OK	2
Q62	Transformer(2)	Tx Support Severely Rusty/Damaged	3
Q62	Transformer(2)	Tx to low < 4.3m to live parts	4
Q63	Isolator/Switch	Isolator/Switch OK	1
Q63	Isolator/Switch	Isolator/Switch Unit Damaged	2
Q63	Isolator/Switch	Isolator/Switch Handle Insulator damaged	4
Q64	Auto Recloser	Auto Recloser OK	1
Q64	Auto Recloser	Recloser Rusty but OK	2
Q64	Auto Recloser	Recloser Severely Corroded	3
Q64	Auto Recloser	Recloser Bushing Damaged/Leaking	4
Q65	Static Balancer/Regulator	Static Balancer/Regulator OK	1
Q65	Static Balancer/Regulator	Corrosion >50% of Surface Area	2
Q65	Static Balancer/Regulator	Bushing Damaged	3
Q65	Static Balancer/Regulator	Oil Leaking from Equipment	4
Q66	HV Fuses	Fuses/Carrier Damaged not OK	3
Q66	HV Fuses	Too low < 4.3m	4
Q66	HV Fuses	Fuse & Carried OK	1

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Q66	HV Fuses	Fuses/Carrier Damaged but OK	2
Q67	LV Fuses	Fuse & Carrier OK	1
Q67	LV Fuses	Fuses/Carrier Damaged but OK	2
Q67	LV Fuses	Fuses/Carrier Damaged not OK	3
Q67	LV Fuses	Too low < 3m	4
Q67	LV Fuses	Fuses/Carrier Damaged but OK	2
Q68	Fuse Box	Fuse Box OK	1
Q68	Fuse Box	Fuse Box Damaged/Decayed	4
Q71	Triggered Spark Gap	TSG OK	1
Q71	Triggered Spark Gap	Evidence of Minor Burning/Arching	2
Q71	Triggered Spark Gap	Evidence of Severe Burning/Arching	3
Q71	Triggered Spark Gap	TSG Broken or Missing	4
Q72	Fault Flow Indicator	Fault Flow Indicator OK	1
Q72	Fault Flow Indicator	FFI Damaged/Broken	4
Q73	Surge Diverter	Surge Diverter OK	1
Q73	Surge Diverter	Evidence of Slight Damage	2
Q73	Surge Diverter	Badly Damaged or Missing	4
Q74	Sealing End	Sealing End OK	1
Q74	Sealing End	SE/Support Slight Damage but OK	2
Q74	Sealing End	SE/Support Badly Damaged/Rusty	3
Q74	Sealing End	Evidence of Oil/Compound Leakage	4
Q75	Conductor Ground Clearance	Ground Clearance In Span OK	1
Q75	Conductor Ground Clearance	Clearance Less than Statutory	4
Q76	Over Road Clearances	No Road/Vehicular Track in Span	1
Q76	Over Road Clearances	Span Crosses Road/Vehicular Track	2
Q77	Over Drive Clearances	No Driveway in Span < 2.5m wide	1
Q77	Over Drive Clearances	Span Crosses Driveway <2.5m wide	2
Q77	Over Drive Clearances	Driveway in span > 2.5m	3
Q77	Over Drive Clearances	No Driveway in this span	4
Q78	Building Clearances (to side)	No Buildings in Clearance Limits	1
Q78	Building Clearances (to side)	Buildings within Clearance Limits	4
Q79	Building Clearances (Under the Line)	No Buildings Under the Line	1
Q79	Building Clearances (Under the Line)	Buildings under Line but OK	2
Q79	Building Clearances (Under the Line)	Buildings under line NOT OK	4
Q79	Building Clearances (Under the Line)	No Span on Pole	5
Q80	Conductor Type	Bare Conductor	1
Q80	Conductor Type	PVC Covered Conductor	2
Q80	Conductor Type	ABC	3
Q80	Conductor Type	VIR Insulated	4
Q80	Conductor Type	Mixed - Bare and Insulated	5

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Q81	Conductor Material	Copper	1
Q81	Conductor Material	Aluminium	2
Q81	Conductor Material	Unknown or Covered Conductor	3
Q82	Conductor Insulation Condition	Conductor Insulation OK	1
Q82	Conductor Insulation Condition	Insulation Damaged/Defective	4
Q82	Conductor Insulation Condition	Not Applicable (bare conductors)	5
Q83	Conductor Regulation	Sag Within Span OK	1
Q83	Conductor Regulation	Sag Within Span Unequal	2
Q83	Conductor Regulation	Conductors out of Regulation	3
Q83	Conductor Regulation	No Span on Pole	4
Q84	Conductor Condition (Inc Earth Wire)	Conductor Condition in Span OK	1
Q84	Conductor Condition (Inc Earth Wire)	Evidence of Conductor Damage in Span	4
Q84	Conductor Condition (Inc Earth Wire)	No Span on Pole	5
Q85	Conductor Attachments	No Attachments in Span	1
Q85	Conductor Attachments	Bird Diverters	2
Q85	Conductor Attachments	Spacers	3
Q85	Conductor Attachments	Foreign Objects	4
Q85	Conductor Attachments	No Span on Pole	5
Q86	Conductor Joints	No Joints in Span	1
Q86	Conductor Joints	1 Joint in Span	2
Q86	Conductor Joints	2 Joints in Span	3
Q86	Conductor Joints	> 2 Joints in Span	4
Q87	Phases with > 2 Joints in Span	< 2 Joints in Any Phase	1
Q87	Phases with > 2 Joints in Span	=> 2 Joints in a any Phase	4
Q87	Phases with > 2 Joints in Span	No Joints In span	5

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Appendix 15 - Feeder condition data assessment template

Feeder Name		No of Phases		Conductor Size / Type	Eg 32mm HDBC	Climate	Normal or Severe	No of Poles in Feeder	
Specification	Eg Heavy Duty Wishbone or BS 1320 etc	Basic Span	Eg 100m	Max Span	Eg 120m	Pole Grade	eg M	Voltage	Eg 11kV
Recommendation	Eg Extensive maintenance, Line strengthening or rebuild/conductor replacement								

Pole or conductor replacement

Pole Changes	Pole No's affected (	Conductor work	Pole No's of spans effected
Due to Pole Condition	Pole Condition (Q21) Item = 3 or 4 Steel Pole Thickness (Q25) = 3 or 4	Due to Conductor Condition	Conductor condition (inc earth wire) (Q84) = 4 Conductor regulation (Q83) = 3 Conductor type (Q80) = 4
Due to Scarf Mark	Scarf mark condition (Q24) = 4 (depth less than 1.5m not acceptable)	Span with > 2 joints per phase	Conductor joints (Q86) = 4 Phases with > 2 joints in span (Q87) = 4
Due to Ground/Building Clearance	Conductor ground clearance (Q75) = 4 Building Clearance - under the line (Q79) = 4 Building clearance - to side (Q78) = 4	Due to Cond. Insulation Issues	Conductor Insulation Condition (Q82) = 4

Additional Work on existing structures

Stay Work Required (Rods, Insulator, Wire)	Wire condition for stay (Q43_1) = 3 or 4 Insulator condition for stay (Q43_2) = 2,3 or 4 Rod condition for stay (Q43_3) = 3 or 4	Cross arm Replacement	Condition of crossarm (Q44_1) = 4
Tree issues	Proximity of trees to pole/span (Q51) = 3 or 4 Proximity of climbable trees by pole/span (Q52) = 2,3 or 4	Pole Leaning	Pole leaning (Q23) = 4
Earthing Issues	Earth – condition of earth (Q40) = 3 or 4 HV earth/OP handle separation (Q41) = 4 Neutral/HV earth separation (Q42) = 4	Cable & cable terminations	Cable condition (Q37) = 4 Cable guard condition (Q38) = 4 Cable Box Condition (Q39) = 3 or 4
Conductor Fittings and Binders	Condition of fittings (Q46) = 4 Condition of binders (Q50) = 2 or 4	ACD's & Signs	ACD Ineffective (Q28) = 4 DOD Condition Q30 = 3 or 4 Property Plate condition (Q32) = 4 Double Circuit (Q34) = 3 or 4 Equipment Name Plate (Q35) = 3 or 4

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Insulator Work Required	<i>Condition of Insulators (Q48) = 2, 3 or 4</i> <i>Insulator type (Q47) = 4</i>	Pole Mounted Equipment	<i>Transformer (Q61) = 3 or 4</i> <i>Transformer (2) (Q62) = 3 or 4</i> <i>Isolator/switch (Q63) = 2 or 4</i> <i>Auto recloser (Q64) = 3 or 4</i> <i>Static balancer/regulator (Q65) = 2, 3 or 4</i> <i>HV Fuses (Q66) = 4</i> <i>LV Fuses (Q67) = 3 or 4</i> <i>Triggered Spark Gap (Q71) = 3 or 4</i> <i>Fault Flow Indicator (Q72) = 4</i> <i>Surge Diverter (Q73) = 4</i> <i>Sealing End (Q74) = 3 or 4</i>
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Notes

- 1) This table shall be used as a template to summarise the condition of a section of line being considered for refurbishment.
- 2) The condition point question numbers relate to the table in Appendix 14 of this document
- 3) To complete the table the pole or span numbers identified from the data shall be entered into the table replacing the questions in italics. As shown in the example cell below It is recommended that the table includes the pole or span effected by the reported issue, together with the condition point question ref and the reported CR code applicable to the component groups in the table. Where necessary these shall be supplemented with appropriate comments.

P1 (Q21) 4, P6 (Q21) 3,

- 4) For each condition point I have entered the level of condition rating or "CR" that is of sufficient severity to require our attention when the line is being considered for refurbishment.
- 5) The content of the form will be used by the re-survey staff to confirm the condition of the initial line report and form the work content included in the subsequent refurbishment scheme. If the resurvey team disagree with the original condition rating the original entry shall be amended and marked with an asterisk to provide an audit trail of the proposed change